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Utilizing Private Spectrum Cellular Network to Motorsport Racing Communication

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In motorsports, especially in car races, communication among the cars, pit crews, and the audience must meet diverse requirements. The racing cars and their drivers communicate extensively with pit crews, who monitor the vehicles to ensure proper mechanical status and driver safety with high-reliability communications. At the same time, broadcasters receive footage from the track to distribute it to the audience worldwide. The current 5G networks allow the creation of private networks, which can be a decent tool to serve motorsport events. Consequently, the devices on the private network can be isolated from the network traffic from the spectators. In this thesis, we plan to quantify and analyze the measurement of using the private and public LTE networks for a racing scenario. The work will involve simulating races in a network simulator. Specifically, mobility traces collected from an actual racing event will be used to simulate mobile nodes as the racing drivers. The fixed location applies to a pit crew, a broadcaster, and an audience, with the base station as an access point. The configuration of base station decides on the type of LTE network in the simulations, we measure each network model and evaluate the performance based on selected metrics. Finally, we discuss the discovery based on our simulations and the potential reasons behind them. ACM Computing Classification System (CCS) Networks → Network performance evaluation → Network simulations Networks → Network performance evaluation → Network performance analysis			
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1 Introduction

This chapter introduces the two subject areas that our work covers, the first is cellular network [1]. The second is motorsport, mainly, we concentrate on car racing on a track like Formula 1. A Samsung research case implemented a 5G network to high-speed vehicles in a disclosed track [2].

1.1 Cellular Network

The cellular network is a group of subnets covering one or several cells as the basic unit, the devices in each cell contain UE (User Equipment) as end users, and eNodeB (Base station) as a central gateway. UEs are typically mobile computing devices such as smartphones or tablets. ENodeB can represent dedicated radio towers or access points installed in a building [3]. However, UEs should retain communication with other UEs even if they move to other cells, and the group of all cells makes a cellular network. Our work scale concentrates only on a single cell with UEs and eNodeB [4].

There are iterative generations of cellular networks with time passing, which contain a series of cellular networks known as 2G, 3G, 4G, and current 5G [5] cellular network. In simulations, we implement a 4G LTE (Long Term Evolution) network in a motorsport scenario. The LTE network enhanced the capability even further by enabling higher speed data rates and lower latency than its processors, the base station in 4G also provides better coverage and transmission power. Moreover, it enables some of the capabilities consumers could never imagine, including video calls and 4k live streaming. We discuss 5G in Chapter 5 with mmWave as a particular version in NS-3, but our work is irrelevant to 2G or 3G networks.

The foundation of our simulations is the 4G LTE networks. The LTE represents a long-term evolution introduced initially in Release 8 [6] and refined in releases 9 and 10 as LTE advanced (LTE-A) [7]. Chapter 2 dives into the general architecture of LTE and explains why private spectrum LTE is beneficial for industrial users compared with public ones. Moreover, the critical reason to select 4G is that the simulation software NS-3 supports LTE [8]. There is also an NS-3 mmWave version [9], but we did not choose it due to its complexity.

1.2 Motorsport

Motorsport can represent the highest specification of the vehicle with the most demanding power and cutting technologies. Modern racing cars intend to have more sensors or actuators, with better connectivity and tractability [10]. The motorsport team functioned as a particular network consisting of racing vehicles and pit crews as user. Furthermore, the application of IoT in motorsports offers the possibility to broadcast high-resolution footage during the racing in real time [10], maintaining sufficient communication quality can be challenging because of the speed of racing cars.

A Case Conducted by Samsung. Samsung has experimented with the combination of the racing car and 5G mmWave [2]. They assigned a 28 GHz spectrum to a tracking device inside a racing car and drove it up to 213 km/h, the measured throughput can still reach 952 Mbps even at the speed of 202 km/h. Which is an impressive result with this level of speed to have high throughput. We also measure throughput by simulating a racing event with two different LTE networks [4].

Samsung also conducted this experiment by monitoring race vehicles and drivers from a remote control room [2]. The footage displayed on screens captured is in real-time, and the pit crews can communicate with drivers in this control room, the drivers wore suits that tracked their health data and transferred them to the control room for safety and rapid rescue. We cannot create a control room to monitor racing cars, instead, the simulations set up one pit crew for each racing car to simulate internal communication and monitoring.

However, that Samsung experiment pushed the racing experience even further for the audience [2]. They can even watch accessible camera footage in real time because of the high throughput from racing cars. The broadcasters can also air high-resolution video with a 5G network, and the technical information of racing cars or drivers is also visible to the audience, which brings an interactive experience that they possibly never had before. Our simulations also create a case where there is a camera in the pit station to capture the footage, a broadcaster to receive and air this footage, and an audience to watch it.

1.3 Hypothesis

We aim to measure and compare the performance of a private LTE network with a public LTE network by simulating a racing event. The assumption is that private LTE networks should perform better than public LTE in specific metrics. The primary reason for this assumption is

that the public LTE network represents a scenario where all the UEs share the same spectrum through one base station, there is a higher possibility of encountering inferences, jitters, or significant delays in shared communication channels. When racing drivers and their pit crews communicate through a private spectrum with an isolated base station, they use a dedicated frequency band and a gateway differentiating from the broadcaster and audience. As a result, they should have more consistent network traffic and lower latency, however, this solution comes with a cost because two base and dedicated spectrum bands can add complexity to the network architecture. We intend to find the difference between private and public LTE networks with a racing event, and what the configuration is so that they can have better performance [11].

1.4 Contribution

We provide a novel use case for the RACECAR dataset [12], which extracts coordinates from one specific scene. Then we make them as the input for the movement pattern of the racing car nodes in simulation, this utilization provides a new use case for their research project.

There is a similar work that implemented communication in high-speed vehicles by using Zigbee [13]. However, they are not focusing on the network performance but only making racing cars tractable. More research documents are needed to cover both cellular networks and racing vehicles. Which ensures the novelty of our work.

The Network Simulator 3 used in this thesis, NS-3.41, has limited instructions to build an LTE model by Python bindings [14]. We provide some referential Python files covering LTE models and make all the simulation scripts and measurements publicly available.

The measurement of our simulations indicates that the private LTE network with shared RAN can not surpass the performance of the public LTE network with the limited capacity in core network [4]. We consulted the NS-3 community, but we are still awaiting a response. In Chapter 5, we speculate possible causes for this behavior, but we could not complete our investigation due to time constraints.

In addition, this document covers five primary research questions: (Table 1-1 Research Questions)

(RQ1) What protocol do pit crews use to communicate with drivers in motorsport racing, and what challenges or constraints do the current solutions face?	Discussed in Chapter 5.
(RQ2) How does a non-private spectrum cellular network (Public LTE) perform compared with a private spectrum one (Private LTE) to achieve high-speed objectives when the network is congested?	Addressed in Chapter 4.
(RQ3) What are the critical metrics of communication during a motorsport race when using a private spectrum cellular network (Private LTE) ? (Example: latency, signal strength, base station coverage, etc.)	Addressed in Chapters 3 and 4.
(RQ4) How can we measure and compare the metrics described in RQ3 between a non-private spectrum (Public LTE) and a private spectrum cellular network (Private LTE) during motorsport racing?	Addressed in Chapter 4.
(RQ5) What are the expected results from experiments implementing private spectrum and non-private spectrum cellular networks in motorsport, and what are the limitations and drawbacks of a private spectrum cellular network (Private LTE) based on these results?	Addressed in Chapters 1 and 5.

Table 1.1: Research Questions. *The first research question is unsolved due to limited availability because motorsport bodies like F1 do not disclose their technical information in communication and broadcasting.*

1.5 Structure of This Document

We introduce the fundamental areas to cover the topics in Chapter 1. Chapter 2 provides an in-depth look at motorsport and cellular networks, with related datasets and network simulation terminologies. Chapter 3 introduces the methodologies of our work, and how to take measurements based on the specific configurations of each network model. The evaluation of the network simulations is in Chapter 4, which lists each metric in public and private LTE networks along with observations. In Chapter 5, we explain our discoveries and limitations. Chapter 6 summarizes all the critical contents from our work and suggests potential future work to optimize the performance in private LTE networks; we also acknowledge the people making this work possible and share all the development files and related datasets.

2 Background

This chapter introduces all the terminology and tools related to our topic. In the first section, we discuss non-private spectrum cellular network configuration, which is part of the public LTE from our simulation. They are relevant to our research, and why might they not be the best option for industrial users. Respectively, the second section is about private spectrum cellular networks as part of private LTE in our work, it has the same aspects of discussions as the public LTE network. The third and fourth sections are the selected simulation software and input dataset for the mobility of our simulations.

2.1 Non-private Spectrum Cellular Network

2.1.1 Consumer Cellular and Topology

LTE is the abbreviation for Long Term Evolution, which is the fourth generation cellular network from 3GPP release 8 in 2009 initially and had an enhanced version in 2011 from release 10 [15]. It has better capacity and antenna support for more sophisticated scenarios to maintain reliable communication and a higher data rate. LTE has yet to disappear from the consumer market even with 5G subscription plans available, it is still active with 6.6 billion users worldwide in 2022 [6]. There is also LTE Advanced (LTE-A) from a later release to enhance 4G capabilities, but we do not discuss or use LTE-A in our simulation.

The general components of an LTE network are UE, eNodeB, MME, SGM, PGW, HSS, and PDN [16]. UE (User Equipment) is a mobile device that consumers use with subscribed cellular network services. Consumers connect to eNodeB (Base Station), which sends and receives all the data from UEs in an assigned cell. The PGW (PDN Gateway) is the intermediate station between PDN (Public Data Network) and SGM, PDN can be a representation of the core network where it has access to global network services. MME (Mobile Management Entry), SGM (Service Group Management), and HSS (Home Subscriber Server) mainly serve to authenticate and verify UEs. However, our simulation models do not configure SGM, MME, and HSS, even though they are part of the Evolved Packet Core (EPC) module in NS-3. (Figure 2-1 Basic 4G (LTE) network topology)

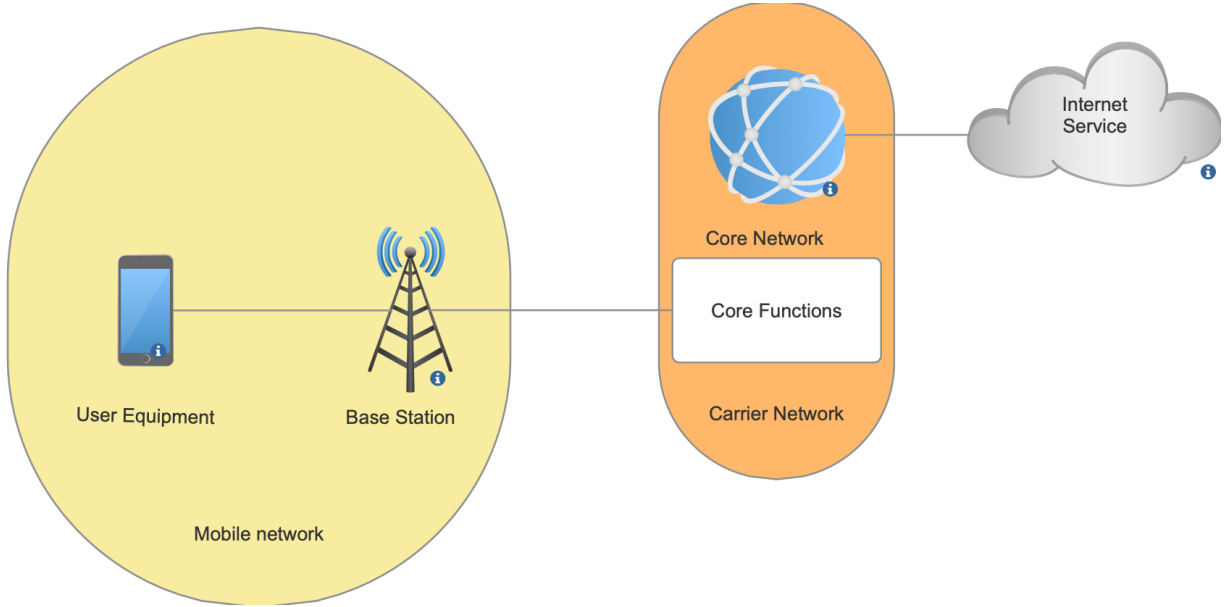


Figure 2.1: Basic 4G (LTE) Network Topology. *The configuration of the core network uses the EPC module in NS-3, which serves all the base stations. In our network models, UEs include racing cars, pit crews, cameras, a broadcaster, and an audience. We use different configurations in the mobile network to measure the performance of each network model.*

Simulation Architecture. There are also two significant terminologies in the LTE network, including Radio Access Network (RAN) containing UE and eNodeB, the SGM, MME, PGW, and PDN are part of Evolved Packet Core (EPC) [17]. In our simulation, we utilize one single RAN for the public LTE network model and two RANs for the private LTE network. We set a remote host to represent the PDN and create a PGM node and SGM to connect the remote host (PDN) to the RAN using an EPC module, NS-3 can also set HSS and MME using EPC helper [8]. The critical difference for each LTE model in our simulations is in the RAN module, while the remaining architecture for all models remains identical.

The communication between UE and eNodeB uses different types of messages, including Master Information Block (MIB), System Information Block (SIB), Radio Source Connection, and more [17]. However, we only apply TCP links between UEs and BSs, and the P2P channel between PDN (Core Network) and RAN in the simulations. Therefore, we do not concentrate on message formats between UEs and eNodeBs, but they can be part of the future work [18]. The simulation application uses TCP Bulksender and TCP OnOff, which use a random starting time and the last timestamp of the RACRCAR data set sampling [12] as the stopping time.

2.1.2 Non-private Spectrum Cellular Network Specification

When it comes to a "Non-private Spectrum Cellular Network", one of the ways to interpret this notion is to operate a primary 4G LTE network (Figure 2-1 Basic 4G (LTE) Network Topology) with a licensed spectrum. In Finland, Telia is one of the mobile carriers that allocate various frequency bands, starting with 3GPP n28 with uplink from 723 MHz to 733 MHz, and 778 MHz to 788 MHz in downlink [19], this frequency range represents that any mobile device supporting this range is accessible to the Telia LTE cellular network services. However, the realistic service takes charge of people who want to use the Telia cellular network, once consumers pay for the subscription, HSS (Home Subscriber Server) and MME (Mobile Management Entry) add this mobile device to their local Telia LTE network. Once these devices can authenticate successfully, they can use the Telia cellular network [16]. The important thing is that anyone with these mobile devices supporting these two frequency ranges should have access to the Telia LTE network after becoming a subscriber. In summary, the Telia LTE cellular network with licensed spectrum is a non-private spectrum cellular network, because any subscriber can use the Telia cellular network through this spectrum. We consider this example as a typical public LTE network.

2.1.3 Limitation of Public LTE Network

The public LTE cellular network allows carriers to have numerous users within each licensed frequency band [19], subscribers can use supportive devices within the carrier frequency range to use their cellular network. However, it is common for many users to access the cellular network through the same base station simultaneously, creating multiple data traffic streams leads to inference, congestion, and high latency. As a result, the extended data queue brings a compromised cellular network performance [20]. The quality of service for carriers can affect whether users want to continue their subscriptions.

The more important thing is that public LTE networks can have compromised security protection. The fact is that consumers' data is only partially on the device when they are using internet services, some third-party companies can collect this data through the videos they watched or shopping lists from their orders in webshops from the cellular network [21]. In this case, if there is a cyberattack on Telia facilities, some users can risk having personal data leaked, which can cause unpredictable disturbances if it happens. For industries with confidential and sensitive data, the consequences of data leakage can be even worse.

2.1.4 Applying Non-Private Spectrum Cellular as Public LTE

For the simulations, we allocate one set of non-private spectrum to the uplink and downlink to one base station in the public LTE model. Therefore, the public LTE network uses only one pair of EARFCN codes [22], which serves all the UEs during the simulation. However, NS-3 cannot support all licensed frequency bands due to the limitation in source files [8], the solution is to select a pair of frequency bands. That is near 723 MHz to 733 MHz in the uplink and 778 MHz to 788 MHz in the downlink. Chapter 3 covers the complete configuration in terms of spectrum.

2.2 Private Spectrum Cellular Network

The previous section introduces the definition of LTE and its licensed spectrum allocation, which serves consumers as public LTE networks [6]. Respectively, there are also private LTE networks, one of the descriptions for private LTE is an isolated subset from the entire LTE architecture [6], which aims to serve a particular area with a group of users. The private LTE network frequently uses shared or dedicated spectrum for communication [18], the application of diverse private LTE networks can benefit industrial users since they are even tailor the configurations of private LTE case by case [18].

The private LTE can operate in numerous architectures by setting hardware configurations [21]. In our simulations, we build private LTE using the reference model of a private cellular network with a shared RAN, and also assign different frequency bands in each RAN to the base station. The network carrier typically allocates the licensed spectrum, especially for a region or based on regulations [19]. The private spectrum frequently utilizes unlicensed or shared spectrum [6], and they can allocate the same bands to all their base stations because it should serve particular groups of users regulated by institutions [23].

2.2.1 Private LTE Network with Industrial Users

The private cellular network can provide an environment isolating a group of UE devices connecting specified base stations [20]. There are also different architectures for configuring a private LTE network, including Stand-Alone Private Deployment, Public-Private Shared RAN Deployment, and Public-Private Shared RAN and Control Plane Deployment [21]. For our simulations, we build the simulation based on deploying a Public-Private Shared RAN, however, we do not consider the control and user planes in the simulations.

The Public-private Shared RAN Deployment is that there is a special eNodeB (Base Station)

connected to public cellular LTE facilities, but it exclusively serves private LTE users [21]. The private LTE network also typically uses unlicensed spectrum or shared frequency bands between eNodeB (Base Station) and UE, the public LTE network has a licensed band for its base station. However, they can operate within the same carrier, which can be a case where Telia runs a public LTE for consumers and private LTE for workers in mining sites.

However, the unlicensed spectrum for private LTE does not indicate that it is accessible and usable for everyone. Some of these shared spectrum is regulated. In Finland, the regulation for unlicensed spectrum belongs to Traficom [23], applying for shared spectrum from Traficom is mandatory if users want to operate a dedicated private LTE network. However, the advantage of a simulation is that it does not allocate the actual spectrum. Therefore, the simulation model selects two spectrums to configure a private LTE base station.

Private LTE has become a commercial product, IPLOOK [24] even provides private LTE service kits for potential customers. They want to implement a private LTE network with a solution once and for all, this private LTE service isolates part of the public operator's network to realize a private-spectrum cellular network environment. The extensive use cases cover manufacturing, mining, transportation, and logistics, expressing that private-spectrum cellular networks or private LTE intends to serve industrial users.

2.2.2 Private Spectrum Cellular Network Specification

The regulation and allocation for private spectrum or unlicensed frequency bands in cellular networks differ in each region. In the US, the shared band of Citizens Broadband Radio Service (CBRS) is the official name for these spectra [6]. In Finland, Traficom (Finnish Transportation and Communication) regulates these bands [23], the private spectrum covers 2300 MHz to 2320 MHz (20MHz) and 24.25 GHz to 25.1 GHz(850 MHz). Any private-spectrum cellular network in Finland must allocate a frequency band in this range for a dedicated local network [23]. However, Traficom does not regulate the hardware specification with its supportive frequency range, the baseline is that building a local network with a private spectrum must apply for permission from Traficom.

2.2.3 Advantage Private LTE

Private LTE can utilize various configurations depending on use cases compared to public LTE cellular networks. We introduce the private LTE with shared RAN deployment in Chapter 2 derived from Public-Private LTE with Shared RAN [21]. Where there is a dedicated group of users isolated from public cellular network services, they should not get heavily

affected when something happens to public base station [21]. Therefore, the traffic in two subnets should be independent, the configuration for each base station can vary depending on particular requirements. The issue is that the core network can still affect both networks, and this condition gets further explanation in Chapters 4 and 5.

As for security, taking a mining scenario as an example, the workers who have their mobile device to communicate should only go through the private base station [18]. Even if a cyberattack happens in the core network (SGM, HSS, PGM, or PDN), it could disturb access to internet service. However, workers can still communicate with each other by just using end devices connecting to private base stations, their data can become safer if stored on the device with encryption. Factually, NS-3 can allow two UEs to communicate via a base station without involving a core network [25] to simulate a case when the core network is down. However, this implementation is possible with C++, not Python bindings in NS-3.41. Our simulation does not cover the security features, but the Chapter 5 discusses security as an essential aspect.

2.2.4 Applying Private Spectrum Cellular as Private LTE

The challenge to simulation a private-spectrum cellular network in NS-3 is that all the available spectrum only covers part of 4G LTE frequency bands [22]. Which represents that the private spectrum in 2300 MHz to 2320 MHz (20MHz) and 24.25 GHz to 25.1 GHz(850 MHz) in Finland [23] is not possible to allocate in simulation. Therefore, the method to allocate a private-spectrum cellular network for racing cars [12] is to use two different frequency bands for each base station, one works as "public cellular network frequency" configured with public base station [22], another serves as an "isolated private local frequency" to the private base station. Both base stations connect to the same core network in LTE architecture as a private LTE network with Shared RAN. (Figure 2-2 Private LTE with Shared RAN Deployment) The best solution to this challenge is to have the support to assign more frequency bands from source files in NS-3 [25].

2.3 Networking Simulator

2.3.1 Network Simulator 3

Networking Simulation 3 is an open source simulator that features discrete events to simulate various Internet environments. It is under GNU GPLv2 license serving for academic research and educational purposes free of charge [8]. The leading developers are from worldwide

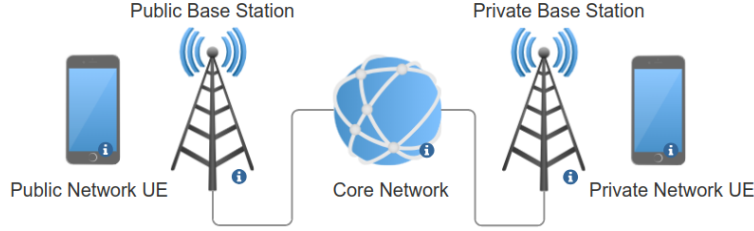


Figure 2.2: Private LTE with Shared RAN Deployment. *The difference from this deployment with Public-Private with shared RAN is that our simulation split two base stations up. It is a simplified version without involving the control plane.*

institutions and communities, NS-3 is actively under development with an incoming release once in a while. We use version 3.41 to build the simulations and take the measurements, released in February 2024, however, NS-3 also has a millimeter wave version [9] to simulate a 5G network with a millimeter wave module. The current version of NS-3 mmWave is 3.38, we do not use the NS-3 mmWave module for running simulations, but we discuss this topic in Chapter 5.

NS-3 provides various documentation for all types of users. The tutorial section covers many basic instructions on how to build and simulate a network in NS-3, it even explains the example files under the NS-3 module path when the simulation enables them [8]. The user manual contains more profound content related to specific functions or objects, and the source files have the definition of functions and modules from the NS-3 model library. We utilize some example files of LTE models to build the network models, additionally, we measure and configure each network's performance based on the flow monitor's instructions and LTE helper chapters [8].

Most documents and model libraries in NS-3 use C++, but we aim to build the simulations with Python bindings API [14]. Which allows users to build programs with the same simulation progress but in a Python environment. All the functions and modules in our document are usable with Python bindings, however, NS-3 claimed that not all the functions in C++ support Python bindings API [8]. The challenge is that LTE models in NS-3 do not have any examples with Python Bindings in NS-3.41, and all our development files and simulation files use Python, which is available in Chapter 6, Section 3.

2.3.2 Reason to Use Network Simulator 3

There are numerous network simulators for researching applications, and we do have an audit for some common options, which is including OPNET [26], OMNET ++ [27], and NS-3 [8]. OPNET officially discontinued the academic version for educational users in 2020,

OMNET++ encountered a library issue during installation on Ubuntu OS. As a result, NS-3 is the only one that does not have technical issues for installation and simulation, moreover, NS-3 is the best option to simulate our network models because of its capability to configure various LTE models.

Moreover, NS-3 is an open-source, free simulator with an active community group to ask questions. Respectively, OPNET only allows subscribers to build simulation programs after the payment, OMNET++ is also a free simulator, but it cannot negatively support Python as NS-3 does [14]. Therefore, if we choose Python over C++ for simulations, which might be faced with a more significant learning curve if OMNET++ is the selected simulator. Furthermore, if there are some questions, the NS-3 user group can provide a platform for users to get consultations compared with other simulators.

The most important reason is that NS-3 enables a wide range of configurations in network typologies and protocols, especially for cellular networks. Because we use an LTE network to provide the communication for all the nodes [12] in NS-3. The driving pattern of the RACECAR dataset is also feasible to simulate by using mobility module [8], the measurement can use a flow monitor, NS-3 provides a modern LTE cellular network topology with many options to configure, measure, and even build a custom module for the LTE module. Although we cannot cover all aspects of LTE models in NS-3 from the simulations, it makes our simulation efficient with some available modules.

2.3.3 How to Simulate a Cellular Network by NS-3

The essential components to operate a simulation in NS-3 contain nodes, net devices, IP address, interface, application, and schedule the operation time [25]. We build the network models based on this principle, and Chapter 3 provides the comprehensive configurations of the simulations. The nodes are end users or devices that are the essential components of a network, they can be racing cars, broadcasters, pit crews, audiences, or base stations. These nodes must connect to corresponding net devices to become usable and serve the simulation with their function [8]. The interface can only be installed with net devices which is also the critical component to get IP addresses.

The links between net devices are channels and major mediums that accommodate data transmission. The typical connection uses a "point-to-point" channel and various attributes [25], this p2p is a classical option for wired connection, there are also wireless ones including Wi-fi and LTE. The IP address needs to be assigned with interfaces for each net device, NS-3 supports IPV4 and IPV6 with various routing protocols[25]. We only apply the IPV4 address to all the interfaces in the simulations, additionally, the internet stack must be installed to

nodes just like smartphones require network access licensed by operators.

The applications are typically serving for generating the traffic during the simulation, the common methods for this module are using UDP or TCP to set client/server or source/sink [14]. The application must be installed with nodes, neither net devices nor interfaces. The UDP typically can generate one or two flows depending on the settings, and the TCP application always has two flows since every packet gets acknowledgment from the receivers. (Figure 2-3 NS-3 LTE Network Diagram for Two UEs) We use TCP as the application of all network models since it is a reliable transmission, and it is simple to justify congested network traffic by duplicate acknowledgment. (Excessive Received Bytes) Once all the configurations are

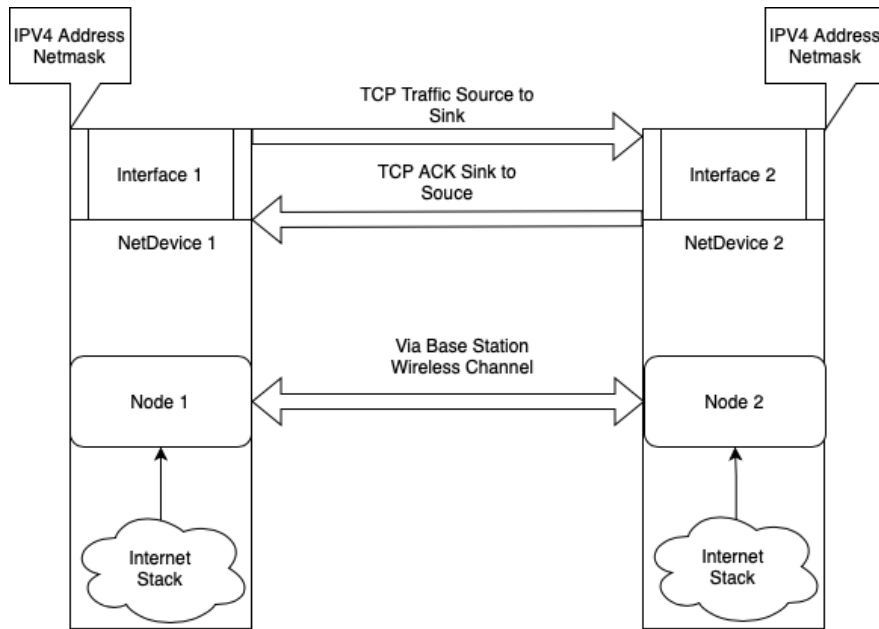


Figure 2.3: NS-3 LTE Network Diagram for Two UEs [25]. Each UE is a node, it must attach to UE net device with UE interface. Once they have IPv4 addresses, they can use a TCP application to transfer the data. The communication channel is wireless in the LTE model via the Base station, and each node must have an internet stack installed. This figure does not include the base station nodes.

finished with nodes, devices, channels, interfaces and applications. The program needs to set the schedule to execute the simulation, it is always a good habit to set a start and end time to avoid infinite simulation [14]. Finally, a destroy function is applied to the program since every execution allocates a memory block and builds some files from the local path, the destroy function can release this RAM by killing the process and cleaning the built files. The important thing is that some configurations must stay before the simulation starts in the initial phase, including setting node, net devices, interface, and mobility. The flow monitor module should be set after the run function because it should collect data during the simulation [8]. We do not cover the reason for this principle rather than focusing how to arrange them to take the required measurements from the simulations.

2.4 Dataset from nuScenes

2.4.1 nuScenes

The nuScene is a public dataset platform containing a large set of autonomous driving datasets to support the research in autonomous vehicles [28]. The development team is Motional, and it contains large groups of traffic datasets from city scales to dedicated racing events. The data is the driven patterns based on GPS coordinates and several aspects, including Lidar, radar, and cameras, which provides extensive attributes from which research teams can choose. We mainly concentrate on the GPS coordinates of RACECAR [12] by filtering out the other attributes.

There is a defined data format for nuScenes, (Table 2-4 Subset of nuScenes Data Schema) the principle to connect each table is the primary token (key), which aims to retrieve the unique data record. Some tables can have another foreign token (key) to lead a data record to other attributes from another table. For instance, the "scene" table has a primary scene token to define a unique scene, which related to its log respectively by using "log token", and the corresponding "logfile", "vehicle" and "location" are available [28]. The valuable data for our simulation is only from a small part of the complete nuScene data schema.

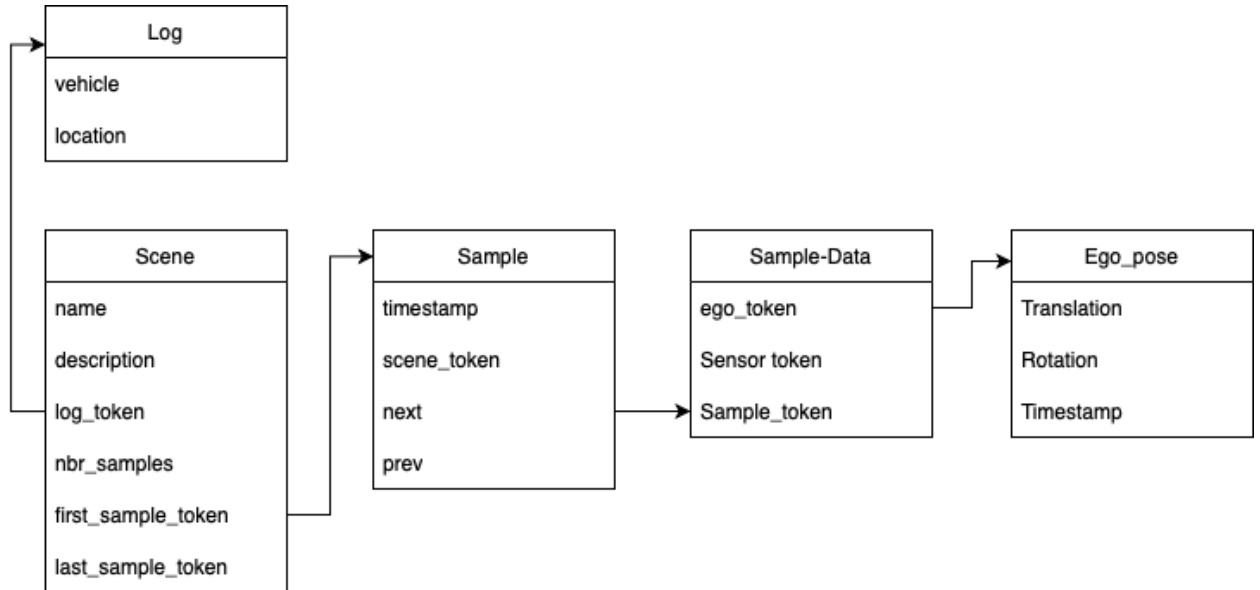


Figure 2.4: Subset of nuScenes Data Schema [28]. *This diagram illustrates only the tables related to the applicable attributes for our thesis work. The starting point is the scene table, which can fetch information from the log table. It also connects the sample and sample data to reach the ego-pose. The coordinates for all the samples within the selected scenes are what the simulation requires for mobility.*

Figure 2-4 shows a subset of the nuScene data schema [28]. The extraction process starts

from the scene because the simulation does not require a complete set of nuScenes datasets for efficient measurement. The complete process and methods are in Chapter 3, the principle is that this nuScenes data format has various tables with different connection methods or tokens. The challenge lies in extracting a dataset that is usable for the simulation of our topic [12]. Figures 2-4 illustrate the selected tables and attributes we must review to reach the data that simulations require.

2.4.2 RACECAR Dataset in nuScenes

The RACERCAR dataset [12] is a set experimented with by an autonomous driving project collaborated with several universities. They hold 27 driving sessions with a total duration of 6.5 hours in two racing tracks, the measurements of these racing sessions are available in nuScene and ROS2 format on the AWS S3 platform. The most valuable of this dataset is that they tested from slow driving with solo vehicles to multiple ones at the highest speed (273 km/h), which can become helpful material for testing LTE networks in simulations with different scenarios. Moreover, even if we only choose one scene that applies to the simulation, it still has 4300 samples from three vehicles that provide sufficient coordinates for the mobility behavior of racing car nodes in the simulations. (Figure 2-5 Partial Data Record of Extract Dataset from the Full RACECAR Dataset)

Node ID	Timestamp	x	y	z
0	0	-173.655977511898	161.846306880886	1.82681509824956
1	0	-244.807501895148	104.856407248306	2.01837131201287
2	0	-159.215047929238	266.113148736405	3.55641805731327
0	0.5	-173.590347061776	161.735333195975	1.82693578347217
1	0.5	-244.960781083428	106.655023195188	2.01869058338526
2	0.5	-191.460153045897	243.430564717425	3.62325680189578
0	1	-171.104731424551	163.742682432996	1.83066234375893
1	1	-247.124946314959	104.327398385209	2.02097072878752
2	1	-202.668903437761	234.483016429559	3.64997092376905
0	1.5	-170.889240408483	161.774069166479	1.83087780263435
1	1.5	-247.683268978444	105.014304662287	2.0250903892534
2	1.5	-218.996871034805	220.447060839644	3.69054509867028
0	2	-171.546053148852	164.065657910502	1.82631075883558
1	2	-244.613419984355	105.858573021276	2.02494630748548
2	2	-229.531012061425	210.720593142947	3.77338880581265

Figure 2.5: Partial Data Record of Extract Dataset from Full RACECAR Dataset. Each "Node ID" is a racing car from its team, the timestamp gets calibrated from (0 second) as the begging of the simulation. X and Y are the coordinates for the racing car in each timestamp. All three cars must be running simultaneously from (0 second) to the end of the simulation

2.5 Summary

This chapter introduces the concept of non-private and private spectrum cellular networks as public and private LTE based on our work. There are three significant deployments for private LTE networks, the one applying to one of our simulation models is private LTE with shared RAN derived from Public-Private Shared RAN network. It has two base stations, one serves public UEs, and the other only communicates with private UEs, but they share the same core network. The public LTE network can have the challenge of fully protecting user's data from security threats, respectively, the private LTE network can maintain the safety of sensitive data and communication even with an unavailable core network. The simulation applies just one spectrum as a "licensed spectrum" to one base station in the public LTE model. As for the private one, there are two base stations with two sets of spectrum to simulate "licensed and unlicensed spectrum" for two groups of subnets in the private LTE model, using the same bands as Telia has is impossible due to the limitation in NS-3 source files.

The simulation software that we use is Network Simulator 3 (NS-3), an open-source free network simulator. We chose this one because alternatives are either unavailable or have technical issues, but the most important thing is that NS-3 allows users to configure and tail the LTE model from this simulator across all the components. However, all the simulations in our work do not involve the NS-3 mmWave module, we contribute to the NS-3 community by building LTE models in Python bindings.

The RACECAR dataset plays a vital role in making our simulations possible. However, it contains fruitful data attributes for each racing car in each scene, we choose to extract the critical elements of the entire dataset to fit our simulations. The simplified dataset version only has one scene with three teams and their location coordinates, they are the input for mobility in racing cars (UEs) in our network models, the procedure of extraction and collaboration is in Chapter 4.

3 Methodology

3.1 Research Method

3.1.1 Quantitative Research

The methodology of our work is quantitative research, which is an approach to conducting and analyzing the experiments in numeral value [29]. Generally, this process requires some statistics techniques to interpret the results of implementation, and the core of this method is that the performance of research work should be measurable. The value of each measurement can justify the performance of each LTE network model [30], moreover, it is crucial to involve statistics for many results from repetitive simulations because it can reflect the trending of performance including minimum, maximum, and median. The plots in Chapter 4 provide the evaluation based on the statistical value of different metrics.

In the hypothesis section of Chapter 1, we claim that the public LTE network should outperform the public one [30]. That should reflect on statistical results not just the direct measurement in metrics. For example, latency is a key performance indicator (KPI), it is not the best option to analyze the latency from one package or just the sum of the delays because they cannot represent the network traffic environment in simulation. The same goes for mean delay, which can even create misleading results because excessive latency (too large or too small) can affect mean delay [31]. Therefore, the evaluation of the measurements utilizes the minimum, maximum, and median to reflect the measurable performance in the selected metrics from each LTE network.

Our results are based on numerous iterations of simulations, ensuring a high degree of reliability and robustness, the experiment's measurements go through with statistical analysis [30]. The total of 16 iterations for 8 in each LTE network is due to simulation hardware can run maximum 4 measurements at once, all the simulations run on a Lenovo laptop with 8-core CPU and 8 GB of memory. Furthermore, the final output uses a separate graph by each LTE network in each metric to illustrate the results to avoid confusion [32], the plots based on our evaluation rely on multiple simulations with different configurations. Therefore, separate plots can be more understandable by categoring each selected metric in each LTE newtork.

3.1.2 Simulation Approach

The method to reach measurable results for our topic is to construct and operate the simulations [33]. Firstly, our simulations aim to measure the selected metrics in two types of cellular networks, the behavior of racing cars in the simulation is more feasible with the available RACECAR dataset regarding efficiency and availability [12]. Secondly, NS-3 (Network Simulator 3) is capable of simulating an environment where it can have base stations that serve its UEs with different configurations. Finally, the flow monitor module in NS-3 can measure multiple metrics based on each flow.

Moreover, our simulations can provide a reference when the LTE network is combined with racing cars [33]. The simulation process also gives a chance to test whether NS-3 can simulate a racing scenario, which also benefits the NS-3 community, as it aims to serve research and education [8]. The complete workflows contain several procedures (Figure 3-1 Simulation Workflow of Thesis Topic), and the following sections introduce each.

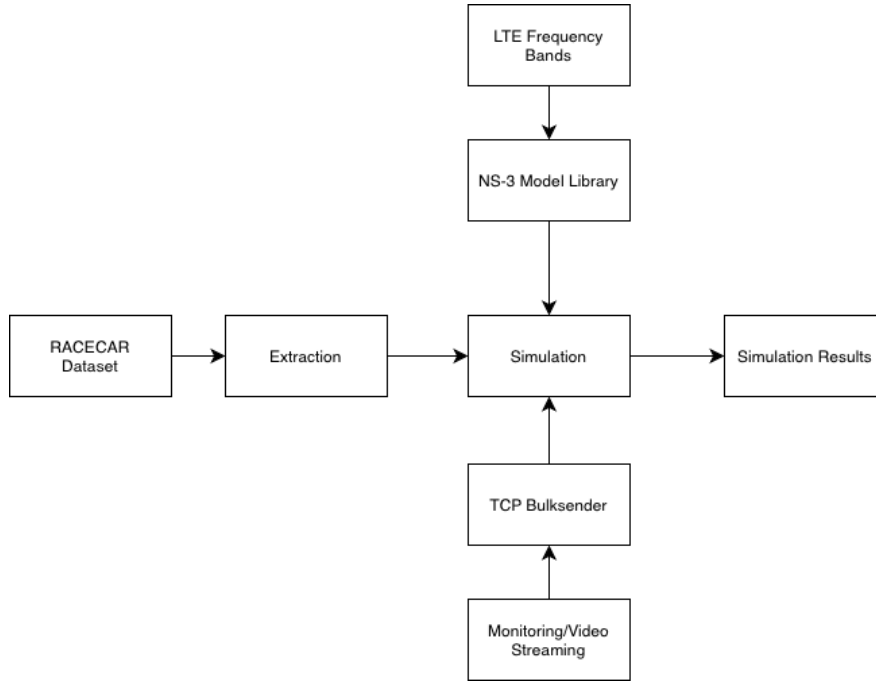


Figure 3.1: Simulation Workflow of Thesis Topic. *The starting point is the RACRCAR dataset, which goes through the extraction process to output a usable CSV file for the simulation. The components of simulation are from LTE bands [22], NS-3 model library [8], and TCP Bulksender [25]. The results of the simulation go with the evaluation process.*

Figure 3-1 illustrates the entire workflow for our work. Which also provides the framework for conducting a simulation according to a research topic. The important thing is that although the RACRCAR dataset provides fruitful aspects of racing cars, the only usable attributes are only a tiny part of it. The simulation requires the LTE bands to configure the uplink and

Multi-Fast-Euro	Multi-Fast-Tum	Multi-Fast-Poli
Track: LVMS	Track: LVMS	Track: LVMS
Vehicle: TUM	Vehicle: Euroracing	Vehicle: TUM
Sample Records: 1469	Vehicle: 1319	Vehicle: 1599

Table 3.1: Scenes in "Multi-AGENT-FAST" Scenario. *There are three teams in "Multi-Agent" scene. The "Multi" represents the dataset that has multiple vehicles for the same features, the "fast" is the fastest speed 273 km/h for all teams, and the "Euro" is the name of the team. Team "Euro" and "Poli" are using the vehicle provided by TUM. Full set in RACECAR repository [12].*

downlink of the base station. We use TCP Bulksender to create a constant data transmission because the simulation aims to set a case where pit crews monitor the racing drivers through their cameras or sensors. The following sections discuss the extraction and configurations of the simulation for each LTE network.

3.2 nuScene Data Preparation

3.2.1 Data Extraction

RACECAR dataset [12] from the nuScenes platform [28] contains 1000 scenes in 22 scenarios with 149186 samples. However, our simulations intend to only partially apply the entire dataset to the simulation due to efficiency and limited resources. In this case, extracting the required part of the dataset is necessary, the selected scenario is "MULTI-AGENT-FAST" including three scenes. (Table 3-1 Scenes in "Multi-AGENT-FAST" Scenario) Including measurements from TUM, (TUM Autonomous Motorsport) EURO (TII EuroRacing), and Poli (PoliMove) with a total of 4467 data records [12].

The extraction objective is to obtain the subsets of the RACECAR dataset with the coordinates of these three racing car locations in each timestamp. However, these attributes are distributed in different tables, RACRCAR dataset follows the format as nuScene defines (Figure 2-4 Subset of nuScenes Data Schema) that scenes do not have coordinates from its table. In this case, extracting the required dataset uses the scene table as the starting point, connecting it with "log", "sample", "sample-data", and "ego-pose" tables. The output should be a CSV file containing the subsets of the full RACRCAR dataset with the selected scenario.

Taking "Multi-Fast-Euro" as the one to explain, this is one of the three scenes under the selected scenario "MULTI-AGENT-FAST". The scene with "Multi-Fast-Euro" is the starting point for retrieving coordinates in the "ego-pose" table, the first step is using this scene token (primary key) to identify the vehicle and location from table "log" as two attributes. The

second step is getting the sample by using the same scene token from "Multi-Fast-Euro" in the scene table to act as a relay, because "next" and "prev" are tokens to identify each sample-data from the sample table that does not have the actual sample but the only token for each sample-data [28].

The most challenging part of the extraction is accessing all sample data because using sample tokens from the "sample" table only is impossible. It must combine with sensor data, if it is applying a sample token to retrieve sample data, users will only get an error message with "key error" [4]. It must use two attributes sample tokens, and the sensor as the second token because this is the only way to retrieve the sample data without errors, once the sample data is accessible through the "sample" table, the last step is to reach the "ego-pose" table, where the most critical information in our simulations is stored. The translation attribute contains the coordinates of the X, Y, and Z, the critical thing is that only the timestamp from "ego-pose" will be part of the export file because this one is related to coordinates. The other timestamp in the "sample" table only represents the time a sample stores but not the moment to capture coordinates. The extraction should generate the datasets covering three racing teams using the sample principle for each team in the "Multi-Fast-Agent" scene.

In other words, the final extraction results should have the vehicle, location from the "log" table, name, description from the "scene" table, and the translation and timestamp of the table "ego-pose" [4]. All the extracted attributes are output as a single CSV file, however, the extracted version of the dataset must go through the processing stage before it is entirely usable by NS-3 simulation. The extraction process can be accomplished on a Jupyter Notebook within a virtual Python environment [4]. Similarly, the data processing stage is also going through with it, NS-3 cannot accept the original timestamp format because the RACRCAR dataset does not use numbers to distinguish the team (Nodes in our simulation). Plus, it has around 32 digits of the number as a timestamp, which is also unusable even when using a long float type. Therefore, the processing stage can make the extracted file usable for NS-3, and it also rearranges the timestamp to ensure NS-3 can load the data from each racing team chronologically.

3.2.2 Data Processing

Although the extracted dataset from RACECAR provides a precious reference to simulate the same movements for nodes in NS-3, it is still unusable before the processing stage because the mobility module from NS-3 assigns the coordinates with identical numbers of nodes. That "Multi-Fast-TUM" cannot be installed in node 1 because index 1 is an integer number [25], not strings, moreover, the translation from "ego-pose" contains X, Y, and Z in one column,

which is more convenient if X, Y, and Z are separable considering NS-3 does not support the Z coordinates. The last thing is that the timestamp from the original dataset is partially chronologically ordered between each scene, Z coordinates can easily lead to an error message from the mobility module in NS-3 [14]. We intend to have three racing cars running simultaneously from the beginning until the end of the simulation.

We must address the incompatibility to solve the discrepancy between the node mobility assignments in NS-3 and the vehicles in RACECAR scenes. NS-3 assigns mobility based on node IDs (getId), whereas the RACECAR dataset only includes vehicle and team names without any corresponding node IDs, the mapping technique can become a solution in that every team in one singular scene takes one number. For instance, the "Multi-Fast-TUM" scene has 1319 samples with corresponding coordinates, which get marked by "0" under a new column "Node ID". In this case, the NS-3 simulation can easily input these 1319 samples to the first node acting as a racing car without adding too many additional lines of code [25]. The sample principle goes to the second team as "1" and the third team as "2". These mapping numbers allow NS-3 to assign coordinates to the corresponding node.

The separation of X, Y, and Z is also crucial for simulation efficiency, the mobility of the NS-3 module cannot support the coordinates in the Z axis [8]. Individual X and Y coordinates in each column are convenient for plotting the movements of three racing nodes in NS-3 to verify that each node takes the correct coordinates from the corresponding team. Jupyter Notebook is the tool to finish this process and the collaboration of the timestamp.

Regarding the timestamp, the positive side is that the RACECAR research team specified that the sampling interval is half a second [12], which allows the processing stage to collaborate the timestamp without creating inconsistency as long as they sample every 0.5 seconds. However, all three scenes use random values as the starting time and increment half a second in NS-3 to avoid burst traffic in the simulation. The ending time remains the same as the last timestamp of each team to ensure the TCP application does not consume the additional network source once the nodes in NS-3 consume all the coordinates in the extracted dataset.

The maximum and minimum of the X and Y coordinates are also helpful because the mobility module requires an appropriate boundary for the movement of the nodes in the simulation. In this case, identifying two bounds from X and Y can ensure a more precise configuration in NS-3 on a proper mobility scale. The output of the processing stage is still a CSV file, and NS-3 can load and assign the data record of each team to the nodes. Using "Node ID" to identify the racing car nodes and "WayPointMobilityModel" [8] as the mobility model. (Figure 3-2 The Process of the RACRCAR Dataset in Extraction and Processing)

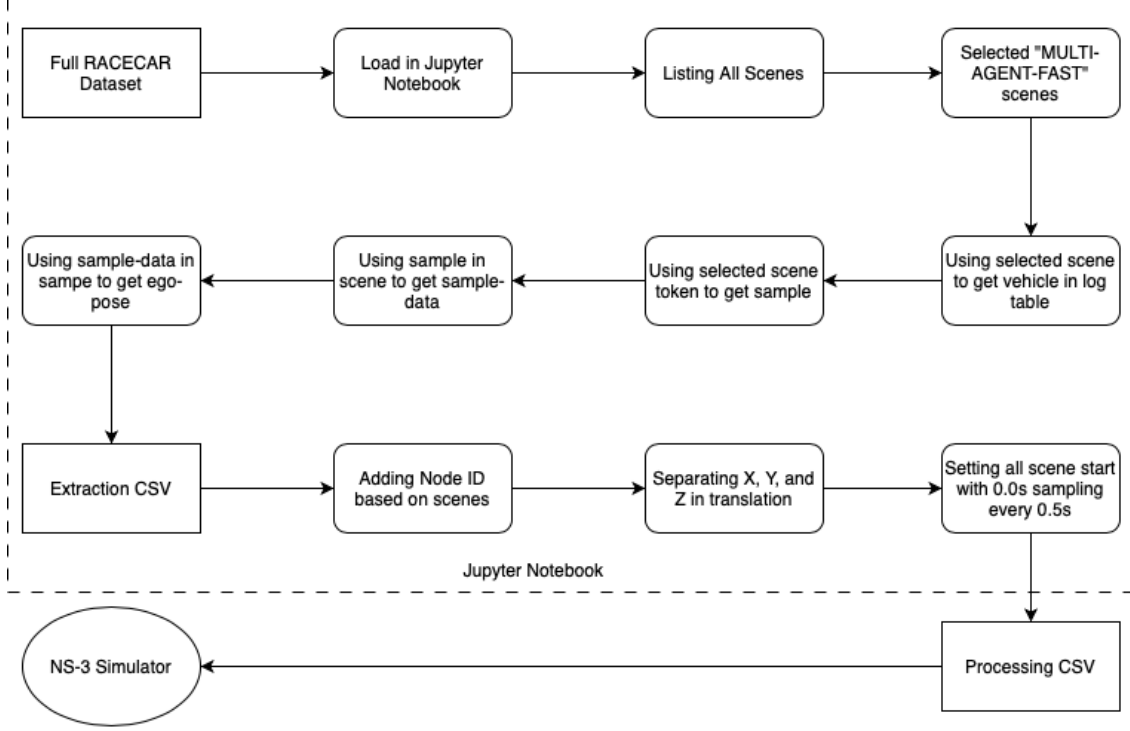


Figure 3.2: The Process of RACECAR Dataset in Extraction and Processing. *The rectangular boxes are CSV files, and the round edge boxes are scripts in Jupyter Notebook. Full scripts including CSV files are available in Chapter 6.*

3.3 Simulation by Network Simulator 3

3.3.1 Function and Mobility of Nodes

In the NS-3 simulator, the specific nodes are the UEs from the perspective of our simulations. The net devices containing UE nodes are UE devices because it is the only method to access a network in NS-3, the base stations are the eNodeB nodes installed in Enb devices [8], which connect to the core network. We concentrate on the roles of each UE node because only they have mobility and applications, and base station nodes work as a gateway for data transmission between UEs. There are two network models for the simulations, including a public LTE network sharing the same frequency band with the non-private spectrum configuration method. A private LTE network isolates racing car nodes with a dedicated base station by using different frequency bands from another base station [4] to project the private-spectrum method as the previous chapters introduce.

Firstly, there are almost two eNodeB nodes (Base Stations) in the private LTE network and only one eNodeB (Base Station) in the public LTE case, but UE nodes remain the same regardless of network models. The first three UE nodes are race cars from selected scenes

in the RACECAR dataset [4], they are using the "WayPointMobilityModel" from the NS-3 mobility module [14], the method to simulate the racing movement is to load the processed CSV file including the coordinates with their timestamp per node. (Figure 3-2 The Process of RACRCAR Dataset in Extraction and Processing) The UE nodes from the 4th to 6th are pit crews in the racing team and use "ConstantMobilityModel" in NS-3 because they are staying in a fixed location. The 7th UE node is the camera that captures the footage from the pit station and transmits it to the broadcaster as the 8th UE node. Finally, the broadcaster is airing this footage to the audience that is the 9th UE node assuming he/she is using a mobile device.

The base station nodes also remain in a fixed location, however, the public base station is located in the center of the track to have a better proportional distance between the racing cars when moving [12]. The RACECAR research team did have a base station close to their locations to collect all the data from racing cars, but they did not record its location, they cannot even recall the approximate location of the track. Therefore, a center location applies to base station nodes in both LTE networks [4].

3.3.2 Cellular Network Service

In our simulations, two types of cellular network service are public LTE and private LTE networks [20]. Chapter 2 introduces the generic implementation for each network model in terms of spectrum. However, this section aims to explain the complete configurations in two cellular networks. It is important to note that NS-3 cannot just assign publicly licensed or shared frequency bands to base stations due to limitations in the source files [25].

Public LTE. The public cellular network service is also the non-private spectrum LTE network mentioned in Chapters 1 and 2. In this cellular network, all UEs are connected to one base station node (eNodeB) with the identical downlink and uplink frequency band (EARFCN code). (Figure 3-3 Public LTE Network Topology for Simulations) Therefore, all UEs in public LTE networks share the mutual base station to send and receive data through one core network [4].

Private LTE. The structure of the private cellar network is the private LTE network with shared RAN [20]. This cellular network uses two base station nodes (eNodeB), which serve as racing car nodes (UE 1, 2, 3), and pit crews (UE 4, 5, 6) as three racing teams. UE 7 acts as a camera in the pit station communicating with the broadcaster (UE 8), that also distributes the footage to an audience (UE 9). (Figure 3-4 Topology of the private LTE

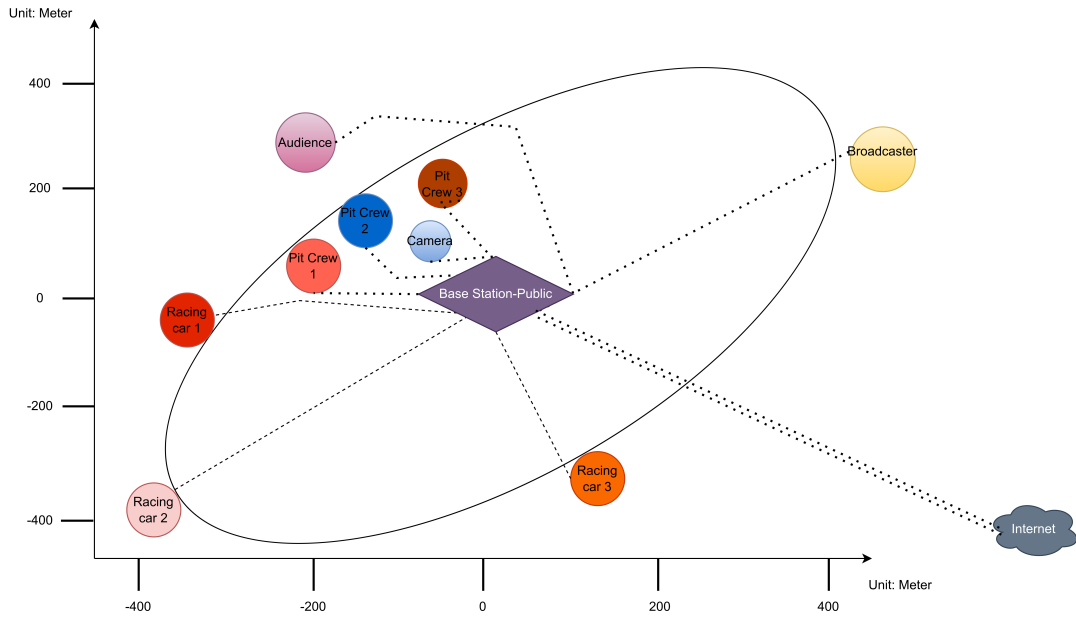


Figure 3.3: Public LTE Network Topology for Simulation. *The racing cars are constantly moving during the simulation with varied speed, the fastest it can reach to 273 km/h. There is only one base station in the center of the track. The audience sits above the track in the lounge area, the pit crews stay in the pit station, and the broadcaster is outside the track in a TV control room. All the nodes except for racing cars stay in a fixed location during the simulation. There is only one internet operator (core network) serving the base station with its UEs.*

network for Simulation) The important thing is that the private base station is connected to race cars and pit crews, however, they can also communicate with the UEs connecting to the public base station because both base stations connect to the same core network. (Figure 3-4 Private LTE Network Topology for Simulation) Our simulations operate TCP applications so that the data transmission between racing cars and pit crews should only go through private base stations, the other UEs should only use public base stations to transmit the data. These networks also use frequency bands (Earfcn code) since Chapters 1 and 2 mentioned private-spectrum cellular networks.

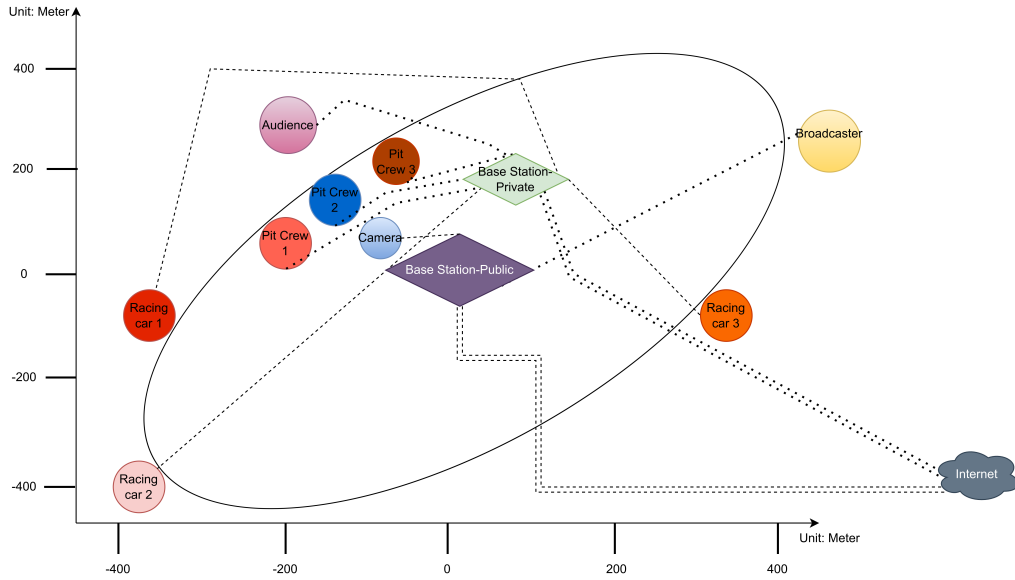


Figure 3.4: Private LTE Network Topology for Simulation. *The racing cars are constantly moving with varied speeds, the pit crews stay in the pit station, the camera captures the footage and transmits it to the broadcaster, the broadcaster streams the video to the audience. The racing cars and pit crew transmit the data through private base stations, and other users use public base stations. All the nodes except for racing cars stay in a fixed location during the simulation. Two base stations are connected to the same internet operator (core network).*

3.3.3 Traffic Patterns

Traffic flows among UEs are the most critical part of generating the measurements in each LTE network, including full-game and lightweight patterns. The former is where all the flows operate with maximum resources across the simulation, the latter set a limited datarate for flows between UEs connecting to the public base station without limiting the datarate of flows between racing cars and their pit crews [4].

Full Game Traffic. The whole game traffic pattern indicates that all UEs in this simulation are constantly transferring data without any constraints [4]. (Table 3-2 Full Game Traffic Pattern) The duration of the simulation is slightly longer than the duration of the last timestamp in the processed dataset with 810 seconds [12]. The first three flows are from three racing cars to their pit crews, referring to the intercom and monitoring the racing drivers from each team. The flow from 4 to 6 is the acknowledgment from the pit crew to racing cars since it uses TCP applications, however, NS-3 does not support one node to be assigned with multiple TCP applications, the solution is to have an intermediate node acting as a camera to transfer the footage from the pit station to the broadcaster. The final flow goes from the broadcaster to the audience, the intention here is to assume that the audience is watching the live stream from their VR devices, although they are just next to the track. This audience is a simplified model unlike people who watch live streams from their homes, because that case is challenging to simulate, considering there are some outdoor and indoor transmissions [10]. The important thing is that these flows are not sequential because the TCP applications start randomly to avoid a massive burst of traffic. However, the plots these display the flows in the described order for better readability.

Senders	Receivers		
	<i>Pit Crew</i>	<i>Broadcaster</i>	<i>Audience</i>
<i>Racing Cars</i>	Full datarate	x	x
<i>Cameras</i>	x	Full datarate	x
<i>Broadcaster</i>	x	x	Full datarate

Table 3.2: Full Game Traffic Pattern. *There are only limited flows between senders and receivers. "x" represents that there is no communication between the senders and receivers. "Full datarate" represents no limitation for datarate for the link between sender and receiver. The following content uses "full" to refer this traffic pattern.*

Lightweight Traffic. The lightweight traffic pattern limits the datarate in the flows among the camera, broadcaster, and audience. The flows between pit crews and racing cars remain unchanged because they always have priorities in either of the LTE networks. However, the initial goal was to use a fixed data packet to simulate a 4k video streaming, but limiting datarate is more reasonable because all the flows should operate at least to the end of the simulation [20]. Therefore, the final decision is to set the datarate as 8 Mbps for flows among the camera, broadcaster, and audience, because this is a level of datarate for transmitting 1080p video with 30 FPS [34]. (Table 3-3 Light Weight Traffic Pattern)

Senders	Receivers		
	<i>Pit Crew</i>	<i>Broadcaster</i>	<i>Audience</i>
<i>Racing Cars</i>	Full datarate	x	x
<i>Cameras</i>	x	Limited datarate	x
<i>Broadcaster</i>	x	x	Limited datarate

Table 3.3: Lightweight Traffic Pattern. *There are only limited flows between senders and receivers. "x" represents that there is no communication between the senders and receivers. "Full datarate" means no constrained datarate for the TCP transmission. "Limited datarate" sets the datarate as 8 Mbps during the whole simulation. The following contents use "light" to refer this traffic pattern.*

Attributes	Values	Explanation
Antenna Model	Isotropic	Equal radiation to all directions.
BaseStation Downlink Bandwidth	25 MHz	Default bandwidth can have the lowest latency and high throughput for downlink.
BaseStation Uplink Bandwidth	25 MHz	Default bandwidth can have the lowest latency and high throughput for uplink.
Path Loss Model	Friis Free Space	The attenuation of signal with distance in an environment without obstacles.

Table 3.4: Configuration Parameters. *The isotropic antenna provides a good reference point for measurement. The default bandwidth is the best for all flows to have the lowest latency and highest throughput after several tests. Friis free space path loss model can create some attenuation for the signal but not cause a large distortion to the transmission to have a massive packet drop.*

3.3.4 Other Parameters

Other aspects affect the measurement including the antenna model, the bandwidth for uplink and downlink, and the path loss model. (Table 3-4 Configuration Parameters) They aim to provide an environment related to realistic LTE network transmission, however, we do not intend to discover the optimal coverage or best configuration for base stations. It is still essential to have these parameters because they provide additional information along with the measurement. The simulation results are based on this configuration, and the following sections explain why they use a particular value/model.

Antenna Model. The simulation’s antenna model is isotropic, allowing the base station to have equal radiation towards all directions [35]. This choice is because the isotropic antenna provides an ideal reference for our simulations, the realistic antenna has unequal radiation depending on the direction. We aim to have a simplified antenna model for base stations to achieve the best possible results in two LTE networks, moreover, the simulation operates in an environment without buildings, obstacles, or indoor conditions. Which isotropic antenna can become a reasonable choice for this setting.

Bandwidth. We concentrate on the uplink and downlink of the base stations in terms of bandwidth because configuring these two links is more accessible to set in the NS-3 module [25]. The result is that setting the same bandwidth as the default value (25 MHz) can have the best performance in terms of delay and throughput. This value comes after attempting with other available bandwidth from smallest 10 MHz to 100 MHz. One of the reasons for this situation is that the default bandwidth for the base station in the source file is 25 MHz, a lower or higher than this value can lead to significantly larger latency and worse throughput due to the possible capacity in the LTE network, we also discuss this issue in Chapter 5. Therefore, 25 MHz is the bandwidth for both links applying to both network models with all traffic patterns [36].

Path Loss Model. The last one is the path loss model since it is a helpful feature to ensure the signal should have some degradation over a longer distance. However, it should be a manageable loss for the packet. The selected model is Friis free space loss model [37] because this loss model servers for a line of sight space environment. In other words, the suitable condition for the Friis free space loss model is where senders and receivers have an unobstructed path to transmit the data, which is the environment for our simulations, because none of the buildings block the transmission between UEs and base stations. Additionally, the other propagation loss model [8] in NS-3 can cause complete packet loss, which does not provide any alternatives compared with the Friis free space loss model.

3.4 Summary

This chapter introduces the quantitative research methodology used to implement the thesis topic. Because the performance among different networks is measurable through numerous values from the simulations [29], it is also the method to make the measurement possible because it is more efficient to simulate the networks with racing scenarios rather than having actual racing vehicles on a physical track.

The method to ensure the RACECAR dataset usable for NS-3 is to extract and process the "Multi-AGENT-FAST" scenario with three teams including TUM, Euro, and Poli. The output file should have the coordinates and timestamps for each team, the starting point of the extraction is from the "scene" table going through "sample", and "sample-data" until "ego-pose" to obtain the translation and timestamp for selected scenes. The "log" table also contributes to the vehicle and track columns for informative purposes. However, the extracted version is unusable for NS-3 before the processing stage.

The processing of the extracted dataset is to add a mapping number to each racing team allowing nodes in NS-3 to utilize the coordinates for each node respectively by using the added column "Node ID". Plus, recalibrating the timestamps using the original sampling interval, as done by the RACECAR research team, can maintain the consistency of the data record and not result in a new dataset. Finally, separating coordinates X, Y, and Z simplifies the process for NS-3 to update the position for the node every half second. The data from the Z axis uses 0 to replace since NS-3.41 mobility does not support it.

The complete configuration for all nodes with their functions, mobility, cellular service patterns, and traffic patterns is made into one table. (Table 3-5 Configuration Table in NS-3 Simulation) There are 11 nodes including 9 UEs, 2 BSs, 2 traffic patterns, 2 cellular networks, and 4 types of mobility.

The principle is that every racing car has a pit crew to communicate with, and this network traffic starts randomly to avoid a burst of packet load. The ending time depends on the last timestamp in the RACECAR dataset for the selected teams in the processed CSV file. The moving nodes are racing cars only, the other nodes stay in a fixed location during the simulation. The difference between the two traffic patterns is the flow among the camera, broadcaster, and audience with their datarate. The full game traffic allows all the flows to use the maximum resource of the network during the whole simulation, and the lightweight traffic limits the datarate for flows among the camera, racing cars, and audience to prioritize the flows in all racing teams.

The other parameters for hardware configuration are also important since they can affect the measurement of the simulation. The selected model for the antenna is isotropic as a reference point. The bandwidth is 25 MHz as the same as the default for downlinks and uplinks of the base stations. The optimal choice for unobstructed communication is to use the Friis free space propagation loss model.

In a word, the simulations should have two LTE models, including public and private LTE. However, each network should have two traffic patterns and four scenarios to simulate. In Chapter 4, selected metrics analyze the measurements from each scenario.

Node ID	Functions	Mobility	Cellular Network	Traffic Pattern
<i>UE-1</i>	Racing Car TUM	RACECAR input	Public & Private	Full Game
<i>UE-2</i>	Racing Car Euro	RACECAR input	Public & Private	Ful Game
<i>UE-3</i>	Racing Car Poli	RACECAR input	Public & Private	Full Game
<i>UE-4</i>	Pit Crew TUM	Fixed at Starting Line	Public & Private	Full Game
<i>UE-5</i>	Pit Crew Euro	Fixed at Starting Line	Public & Private	Full Game
<i>UE-6</i>	Pit Crew Poli	Fixed at Starting Line	Public & Private	Full Game
<i>UE-7</i>	Pit Station Camera	Fixed at Starting Line	Public & Private	Full Game & Light Weight
<i>UE-8</i>	Broadcaster	Fixed Position	Public	Full Game & Light Weight
<i>UE-9</i>	Audience	Fixed Position	Public	Full Game & Light Weight
<i>Enb-1</i>	Public LTE Base Station	Fixed Center Track	Public & Private	Full Game
<i>Enb-2</i>	Private LTE Base Station	Fixed at Starting Line	Private	Full Game

Table 3.5: Configuration Table in NS-3 Simulation. *RACARCAR input* represents the position of the nodes that get updated every half second in X and Y coordinates from the extracted and processed dataset. *Fixed at the starting line* represents the pit crews stay where racing cars departure statically, the coordinates of fixed position for broadcaster and audience are based on the layout from tack in real life [38], fixed central is the approximate center position of the track [12]. *Full Game* represents that the UE does not any limited datarate in full traffic pattern, *Lightweight* marks the UEs have limited datarate in light traffic pattern.

4 Evaluation of Simulation

4.1 Critical Metrics for Evaluation

4.1.1 Flow Monitor Module

The flow monitor is a part of the modules that track packets in the transmission layer that support TCP and UDP protocols in NS-3.41 [39]. The principle of monitoring flows is to add tags for transmitted packets in the network. This module allows users to track and classify specific metrics based on single or multiple flows. However, this module can only support IPv4 and IPv6 classifiers and unicast with the NS-3.41 version [39].

Moreover, detecting lost packages in the flow monitor module can not fully represent the loss. The primary method for the flow monitor to justify if a packet gets lost is to wait for tagged packets for 10 seconds, any untraceable packets longer than that are "lost", in another case, the packets are in a queue but the simulation ends, then these packets are "lost". The second case is not the accurate justification for loss because the simulation eliminates the queuing before these packets can be transmitted. Therefore, we set a margin period between the end of the simulation and the stopping time of the TCP applications to allow the packets in a queue to get transmitted. Our simulation generates a large volume of data from 10 links [39].

Compared with other tracing methods including the Pcap, Ascii, or Trackback function [39] in NS-3, the flow monitor provides a simplified solution to track and observe the flows from the networks. Pcap, Ascii, and Tracback concentrate more on the single net device node or interface. This chapter evaluates private and public LTE network performance based on flows other than a particular net device or interface. Therefore, the flow monitor is a reasonable option to choose.

4.1.2 Selected Metrics

There are numerous metrics available [39] in the flow monitor module, but we select only part of them to evaluate the performance for both networks. This chapter also explains the reason to select these metrics. (Table 4-1 Selected Metrics and Descriptions)

The first four metrics help verify if TCP applications operate as scheduled to avoid inconsistency due to timestamp issues. The txBytes and rxBytes are essential parameters to calculate

Available Metrics	Description
<i>timeFirstTxPacket</i>	The first timestamp when packet transfers.
<i>timeFirstRxPacket</i>	The first timestamp when packet receives.
<i>timeLastTxPacket</i>	The final timestamp when packet transfers.
<i>timeLastRxPacket</i>	The final timestamp when packet receives.
<i>txBytes</i>	The total bytes for transmitted data.
<i>rxBytes</i>	The total bytes for received data.
<i>txPackets</i>	The number of transmitted packets.
<i>rxPackets</i>	The number of received packets.
<i>lostPackets</i>	The number of lost packets.
<i>delayHistogram</i>	The histogram of delay including index, width, and number of transmitted packets during a single width.

Table 4.1: Selected Metrics and Descriptions. *The first four metrics are timestamps to ensure consistency with the scheduled time in simulations. The txBytes and rxBytes are elements used to calculate the throughput and goodput. The delayHistogram is the input data to plot latency distribution. The lostPackets is for monitoring purposes.*

the throughput and goodput to evaluate the measurement of the simulation. The number of txPackets, rxPackets, and lostPackets are the reference values to ensure regular traffic in each flow. Because the minimal number represents the simulation encountering some issues with network traffic, finally, the delayHistogram is one of the critical sections for generating the plots of latency distribution by each flow. The delayHistogram can be a reference [10] for its Quality of Service (QoS) and Quality of Experience (QoE) [40]. However, our evaluation concentrates only on how to utilize it to the required plots.

Latency and throughput are part of the integrity KPI (Key Performance Indicator) for the LTE network [40], which focuses primarily on the experience of end users using the network. We define racing cars, pit crews, and the audience as the users of the LTE networks, who might have a positive or negative experience based on the measurements of these indicators. Specifically, the evaluation prioritizes the communications of racing cars and pit crews other than the other flows.

There are three files to store measured data, one is the ".xml" file from the NS-3 flow monitor module [8], and two other ".csv" files are from the two functions we made by Python bindings. One of them captures the throughput and the goodput of TCP flows in the simulation, and another one contains the start and counts from the delayhistogram of TCP flows. These CSV files are just a subset of the ".xml" file, it is more efficient to have two separate files as the input for plotting rather than making all the plots only based on ".xml" files.

To utilize X-axis space efficiently for each plot, the transmission of each flow uses the abbreviation of each flow with its UEs. The full description is in Table 4-2.

Flow	Description
<i>RT1</i>	Racing Car 1 → Pit Crew 1 with monitoring in video or audio communication by TCP applications.
<i>RT2</i>	Racing Car 2 → Pit Crew 2 with monitoring in video or audio communication by TCP applications.
<i>RT3</i>	Racing Car 3 → Pit Crew 3 with monitoring in video or audio communication by TCP applications.
<i>CB Footage</i>	Camera → Broadcaster with the transmission of footage by TCP applications.
<i>BA Streaming</i>	Broadcaster → Audience with live streaming service by TCP applications.

Table 4.2: Flow Descriptions. *Every flow has its ACK (Acknowledgement) response because the links run on TCP applications. However, ACK flows does not appear in the evaluation plots. All the observations use these abbreviations to specify certain flows in the following contents.*

4.2 Throughput

The way to obtain throughput is using txBytes divides the difference between the timeLastTxPacket (Last Timestamp Transmitted Packet) and timeFirstTxPacket (First Timestamp Transmitted Packet) [39]. (4.1) However, this metric can only reflect the volume of data that UEs send out during the transmission period. To measure the accuracy of data that arrives at the destination UEs, it should be goodput [41]. (4.2)

$$Throughput = txBytes / (timeLastTxPacket - timeFirstTxPacket) \quad (4.1)$$

4.2.1 Public LTE

The measurement of the public LTE network includes 3 racing cars, 3 pit crews, 1 camera, 1 broadcaster, and 1 audience communicating via only one base station. There are a total of 16 iterations, including 8 iterations in each traffic pattern. Each simulation runs up to 810 second to ensure that all the racing cars utilize all the coordinates from the processed RACECAR CSV file, and the queue can clear out all the packets [12]. (Figure 4-1 Throughput plots in Public LTE)

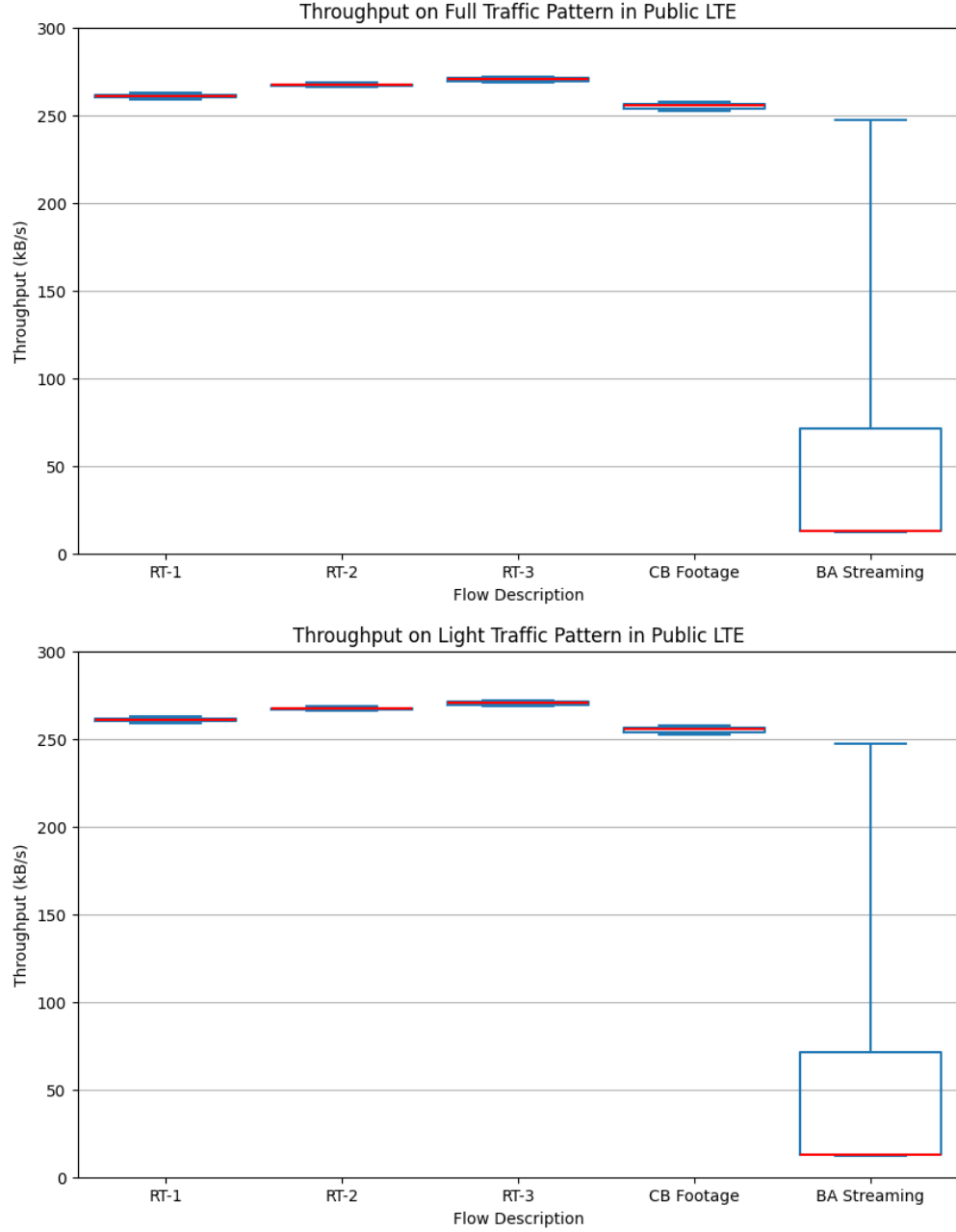


Figure 4.1: Throughput Plots in Public LTE. The X axis is each flow between two UEs. All is connecting to one base station. The Y axis is the range of throughput in kB/s. The boxes have 25th and 75th of the percentile of data as the bottom and top, the red line is the median value. The range of the error bars represents the minimum and maximum throughput of each flow.

4.2.2 Key Observation

Figure 4-1 illustrates the throughput trend in public LTE networks with two traffic patterns. The upper plot is the result based on the full traffic pattern in the simulation. The boxes for each flow represent the range of 25th and 75th of the percentile in throughput, the maximum throughput is the upper bound of the error bar, and the minimum one is on the bottom line of the error bar by each flow. Each box has a red line highlighting the median throughput. The throughput range for most of the flows is between 250 kB/s and 300 kB/s except for the last flow BA streaming. The range of the error bars is narrower for the first 4 flows in the plot. That represents that these flows' throughput is primarily consistent except for the last flow because it has the longest error bar.

The lower plot is the measurement of the throughput of the light traffic pattern. That limits the datarate between the last two links (CB Footage and BA Streaming) to allow racing cars and pit crews to have more bandwidth to transfer the data. However, the median of throughput as red lines for flows between racing cars and pit crews (RT-1,2,3) is only slightly higher than the full traffic pattern close to 300 kB/s. It does reflect on the plot, but the measured throughput in light traffic is higher from CSV files storing selected metrics by each flow. The error bars of the first four flows are narrower than the last two, indicating these flows remain consistently around 280 kB/s during the simulation. However, the last flow in BA streaming is also an exception in low traffic patterns with the longest error bars and lowest median in throughput plot.

The most important thing is that the limited datarate is 8 Mbps, which is equal to 1000 kB/s as the setting for CB Footage and BA streaming in the light traffic pattern. However, even the maximum of the upper bound of the error bar only reaches around 220 kB/s from these flows. The reason for this is that the only method to set certain datarates for specific links is using OnOff application [8] for TCP in NS-3, and the default setting with On (Transmitting Data) for 1 second and Off (Not Transmitting) for 1 second. Therefore, the mechanism to transfer data in the last two flows is taking turns by ON and Off states within 1 second interval. That means these two flows do not constantly transmit the data during the simulation period. Moreover, the simulation sets the path loss model where the signal gets attenuated over the distance, and the acknowledgment consumes part of the bandwidth for these flows. However, the plots do not conclude them. As a result, the final throughput is lower than the setting value because these factors can downgrade the signal strength leading to lower throughput.

4.2.3 Private LTE

The private LTE network uses two base stations with the same number of UEs as public LTE. However, the private LTE measures two different locations of the private base station with 4 iterations. Which represents that every traffic pattern still has 8 iterations, and the total simulation times remain at 16 as the same as the public has. (Figure 4-2 Throughput Plots in Private LTE)

4.2.4 Key Observation

Figure 4-2 shows the trending of throughput in private LTE networks from the simulation with two traffic patterns. The upper plot is the result of the full traffic pattern. The last two flows CB footage and BA streaming have higher throughput than the flows RT-1, RT-2, and RT-3. The median throughput is the red lines staying between 100 kB/s and 150 kB/s for the first three flows from three racing teams that connect to the private base station. The rest of the flows CB footage and BA streaming are connected to the public base station and have a median throughput between 150 kB/s and 200 kB/s. The upper and lower bounds of the error bars are narrower with the first three flows from the private base station compared with the last two from the public base station. Therefore, the overall throughput in a full traffic pattern has a stable throughput range for all flows, except for flow BA streaming. One of the potential reasons is fluctuation or congested traffic in the network because a full traffic pattern does not limit any flows by any means, it might also be the exact reason for the flow of BA streaming in the public LTE network.

The lower figure is the light traffic pattern in private LTE network simulation. The constraint datarate remains the same as public LTE in light traffic as 8 Mbps, which is equal to 1000 kB/s for the flows CB footage and BA streaming. The plot shows that the throughput in flows RT-1, RT-2, and RT-3 are slightly improved but still between 100 kB/s and 150 kB/s with the red lines as the median throughput. However, the difference between the two traffic patterns in private LTE is that the flows of CB footage and BA streaming have lower throughput in light traffic patterns. Similarly, this result is due to the feature of OnOff TCP applications, with the default setting as 1 second interval to switch On and Off between transmission and idle. The path loss module can also affect the signal when the distance increases in two UEs. Therefore, the final throughput from the flows with limited datarate is lower than the setting value 1000 kB/s. However, unlike the full traffic pattern in the last flow of BA streaming, the light traffic plot does not have an extremely long error bar and low throughput. Which indicates that all the flows remain in a stable range of throughput consistently.

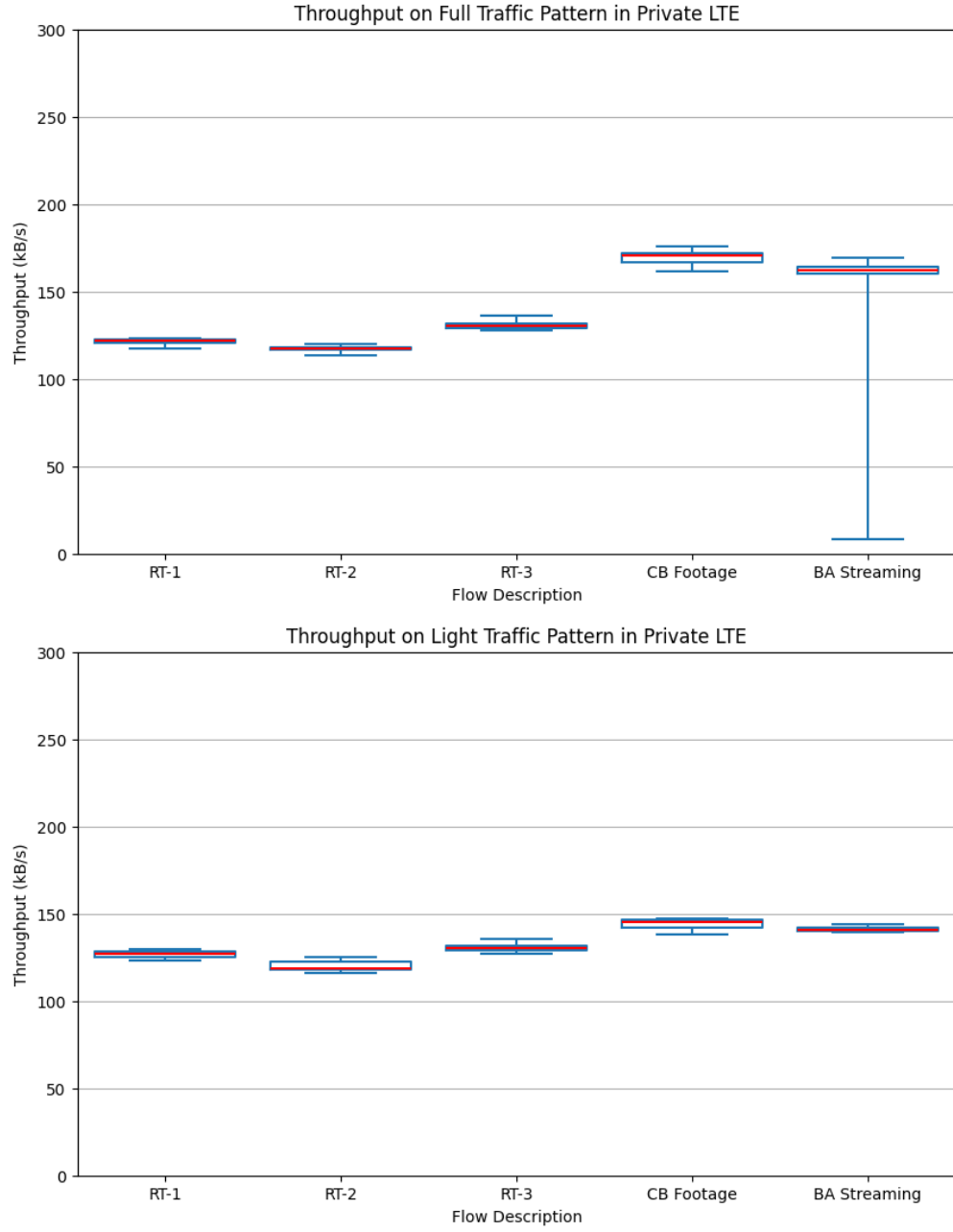


Figure 4.2: Throughput Plots in Private LTE. The X axis is the flow between every two UEs. The flows of RT-1, RT-2, and RT-3 are connecting to private base station, the rest of flows is with public base station. The Y axis is the range of throughput in kB/s. The boxes have 25th and 75th of the percentile of data as the bottom and top, the red line is the median value. The error bar represents the minimum and maximum throughput of each flow.

4.3 Goodput

The goodput is essential to evaluate the volume of valuable data arriving at the destinations. It can reflect the quality of service (QoS) in each flow during the simulation [41]. The method to attain goodput is to use *rxBytes* to divide the difference between *timeLastRxPacket* and *timeFirstRxPacket*. (4.2)

$$Goodput = rxBytes / (timeLastRxPacket - timeFirstRxPacket)$$

4.3.1 Key Observation

In general, the goodput of private and public LTE in full and light traffic patterns remains identical to the throughput plots in Figures 4-1 and 4-2. Therefore, this section does not use any plots because all the trends of goodput by each flow do not have visual differences from throughput ones. However, the first thing is that each flow does have lost packets, but various factors affect the number of lost packets. It is not our primary intention to find the most influential factor that causes lost packets. The second thing is that the same trending from goodput plots can emphasize that TCP can provide reliable communication because most transmitted data arrives at the destination correctly. Some packets may have retransmission due to congested network traffic or other conditions. We are aware that TCP retransmissions can exist. However, excessive retransmission can lead to higher goodput than throughput because the flow monitor cannot distinguish the packet as the first transmission. Fortunately, we have yet to have this situation in our results.

4.4 Latency

The method to evaluate the latency in all simulations is to utilize the delay histogram in each flow from the flow monitor module in NS-3 [4]. The latency distribution is based on each flow and the counts of the packets in a specific range of delay [42]. The plots for latency distribution (Figure 4-3 Latency Distribution in Public LTE, Figure 4-4 Latency Distribution in Private LTE) cover the data with both traffic patterns by each network. Because neither of the traffic patterns in each LTE network shows significant differences in latency. The plots use blue, red, and green markers for the minimum, mode, and maximum of the delay value in each flow. The reason for not using the sum of delay is that the number of packets transmitted for two LTE networks is unequal. Therefore, when the number of packets for an LTE network is higher, a more considerable delay can accumulate, but it cannot reflect the

network traffic in this way. The latency distribution plot can show the delay range for each flow and the delay most of the packets have by using these statistical markers [43].

4.4.1 Public LTE

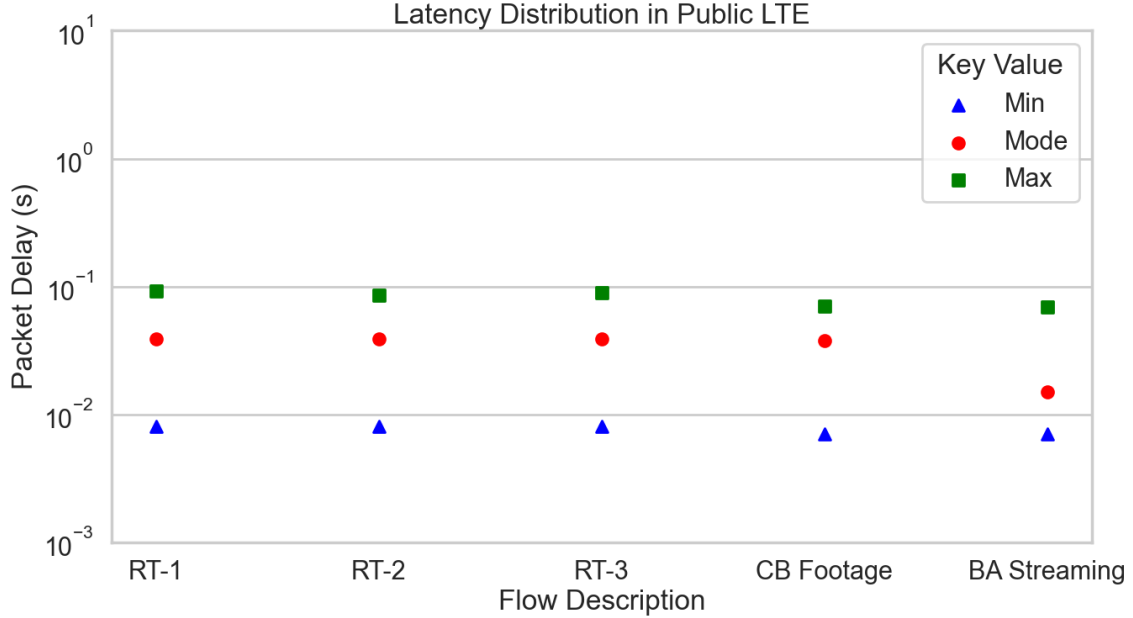


Figure 4.3: Latency Distribution in Public LTE. The X axis is the flow between every two UEs, all connecting to one base station in the Public LTE network. Using a logarithmic scale, the Y axis represents the range of delay in the simulation. The blue triangle, red circle, and green square markers are the minimum of the delay, the mode of delay based on the count of the packets and the maximum delay by each flow. The latency distribution of the public LTE network is using scale 0.001 second to 10 seconds.

4.4.2 Key Observation

All the latency distribution plots do not separate the traffic patterns. Because the measurements from both traffic patterns are identical in each LTE network. All of these flows transmit the data with latency below 0.1 second in a public LTE network. The mode markers represent the delay with the most counts around 0.04 seconds from flows RT-1, RT-2, RT-3, and the flow in CB footage. The flow BA streaming has its mode delay around 0.01 second. The maximum delay appears in the first flow RT-1 around 0.09 seconds, and the minimum delay for all flows is close to each other. The lowest mode delay is on the last flow BA streaming below 0.02 seconds, the rest of the flow has all their mode delay near 0.04 seconds indicating most of the packets have this delay consistently.

4.4.3 Private LTE

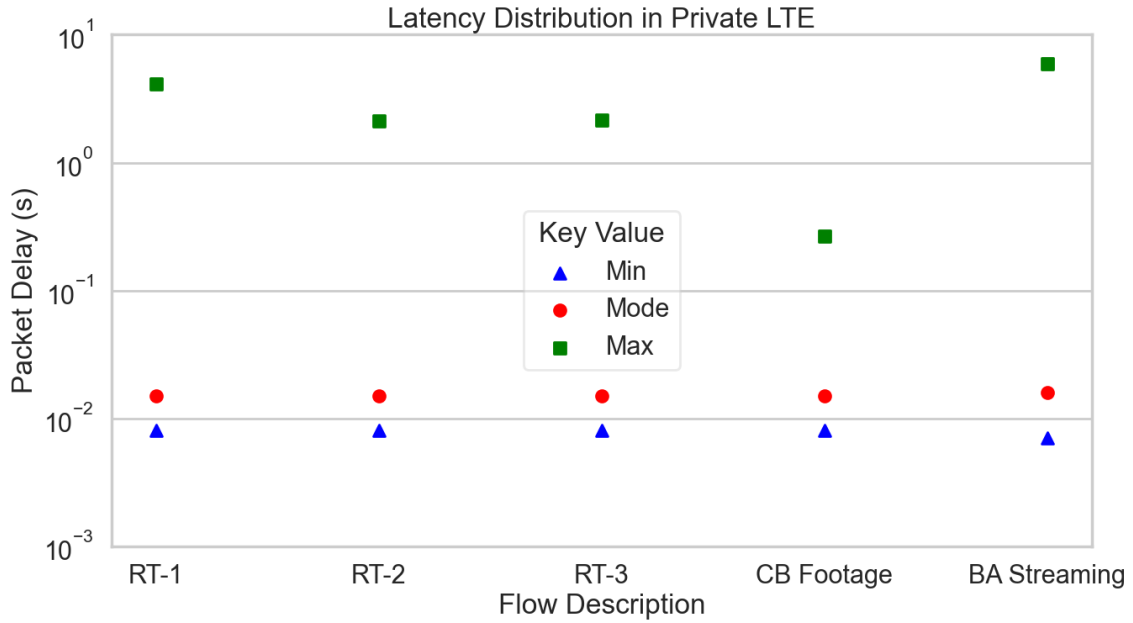


Figure 4.4: Latency Distribution in Private LTE. The X axis is the flow between every two UEs, the flows of RT-1, RT-2, and RT-3 are connecting to private base station, the rest of flows are with public base station. The Y axis represents the delay range in the simulation with a logarithmic scale. The blue triangle, red circle, and green square markers are the minimum of the delay, the mode of delay based on the count of the packets and the maximum delay by each flow. The latency distribution of private LTE network is using a scale between 0.001 second to 10 seconds.

4.4.4 Key Observation

The plot for latency distribution for private LTE shows significant differences compared with the public LTE. The latency distribution plot in private LTE uses the same scale within 0.001 second to 10 seconds in the Y axis by just applying logarithmic. It also does not separate the traffic pattern since they have the identical latency distribution. The flows RT-1, RT-2, and RT-3 connecting to the private base station have the maximum delay of the packet around 2 seconds to 4.5 seconds. The flows CB footage and BA streaming connecting to the private base station has maximum delay from 0.3 seconds to 6 seconds. All the flows consistently have their mode delay below 0.5 seconds. However, the private LTE has a larger maximum delay than the public case. There is a further discussion in the next section of this Chapter and Chapter 5 about this condition. Private LTE generally shows that packets are distributed mainly within 1 seconds delay. However, some packets have larger delays in each flow since two base stations share the capacity of one core network.

4.5 Explanation for Different Results

4.5.1 Throughput

The plots for the private LTE network with their throughput (Figure 4-2 Throughput Plots in Private LTE) indicate that the trending of the throughput of each flow is always lower than the public LTE. Although this result is different from the hypothesis, the potential reason for this phenomenon is the capacity of the LTE network in NS-3. To address this claim, we conduct a "bare LTE networks" experiment [4]. (Figure 4-5 Bare LTE Network Configurations) One network as a public bare LTE contains 4 UEs, 1 base station, and 1 core network. Another simulation has the same amount of UEs, but base station 1 connects UE 1 and 2, and base station 2 with UE 3 and UE 4. The position of the base station remains the same as in the simulation of the racing scenario, but all the nodes stay in a fixed location during the simulation. Because this experiment aims to find if the capacity of the network is the factor creating the different throughput and latency distributions by excluding other factors, the duration of this "bare LTE network" runs for 20 seconds.

Unsurprisingly, the throughput of bare LTE network simulations shows the same trend as our simulation from the racing scenario. (Table 4-3 Throughput in Bare LTE Networks)

Sender->Receiver	Bare LTE Network	Throughput (kB/s)
UE1-UE2	Private	90.284090
UE3-UE4	Private	120.675088
UE1-UE2	Public	302.303333
UE3-UE4	Public	302.167677

Table 4.3: Throughput in Bare LTE Networks. *This measurement is based on two simulations, each with 4 UEs. The private bare LTE network has base station 1 covering UE 1 and 2, base station 2 connects UE 3 and 4, and the public bare LTE network has only one base station for all UEs. All the flows run on TCP Bulk sender application with unlimited datarate.*

Tables 4-3 indicate that all flows with their throughput are lower in private bare LTE networks because they use two base stations to split the core network's capacity. However, it is insufficient to claim that the capacity for the flow between the core network and base station is around 604 kB/s. (Sum of the throughput from half of the flows in an LTE network) Because many factors affect the throughput even with this bare LTE experiment. Which is the reason the simulation of the racing scenario is even lower because the configuration adds moving UEs [12], the path loss model [37], and the isotropic antenna model [35]. Therefore, these factors can create additional inference and attenuation of signals leading to even lower

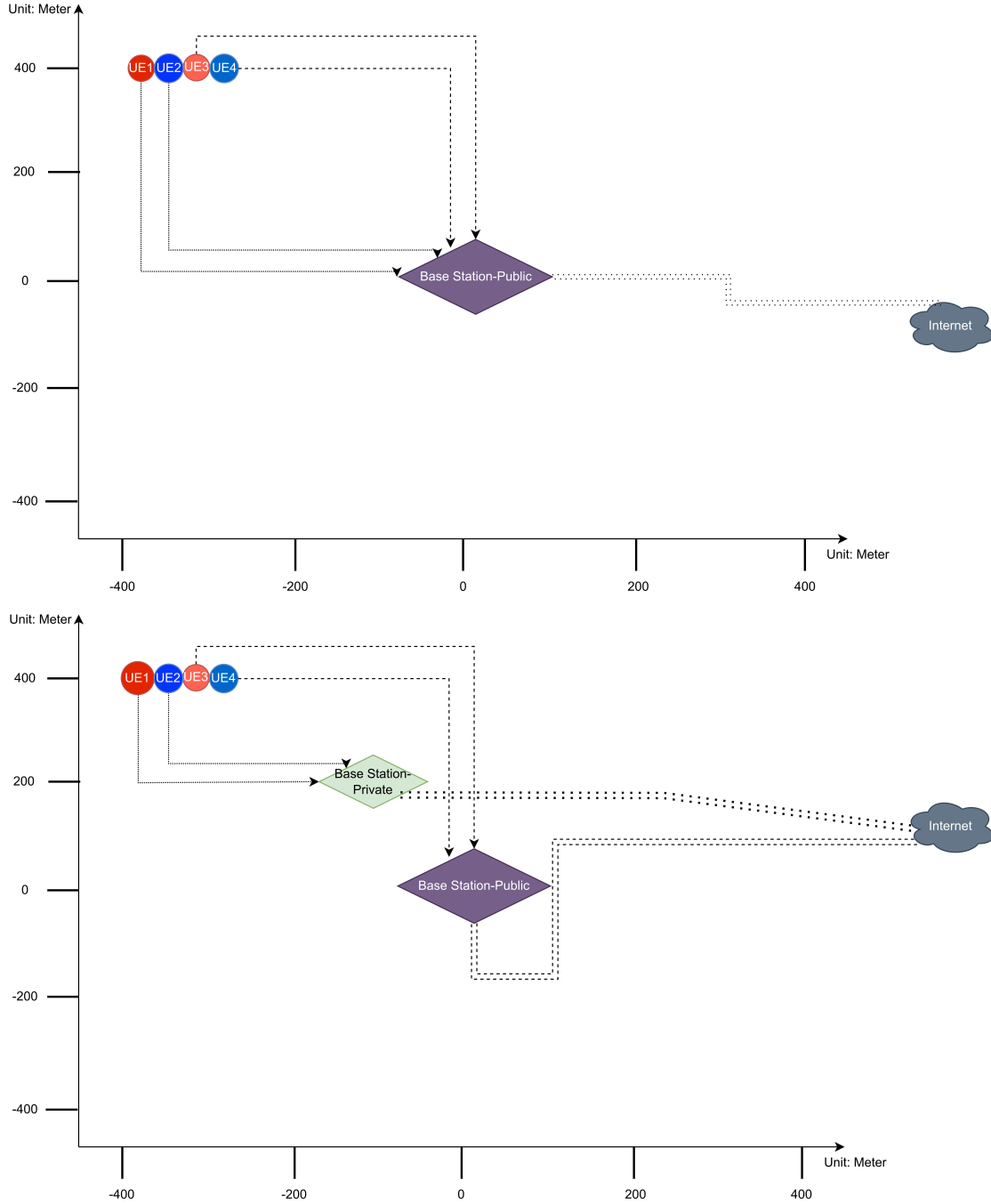


Figure 4.5: Bare LTE Networks Configurations. The upper figure is the public bare LTE network simulation with 4 UEs using TCP applications through just one base station. The lower figure is the private bare LTE network simulation with the same TCP applications, UE1 and UE2 connect to a private base station and UE3 and UE4 connect to a public base station. All the nodes stay in a fixed location during the simulation, the location of the internet (core network) in the figure does not indicate its actual position because it is unknown in NS-3.

throughput. Furthermore, the experiment with the bare LTE network leaves a clue about why the default bandwidth is a better setting because a larger bandwidth might overload the capacity of the core network leading to lower throughput and larger delays.

4.5.2 Delay

The "bare LTE network" also uses the delayhistogram from flow monitor module [39] to generate the latency distribution plots by each flow. (Figure 4-6 Latency Distribution in Bare LTE Network Comparison)

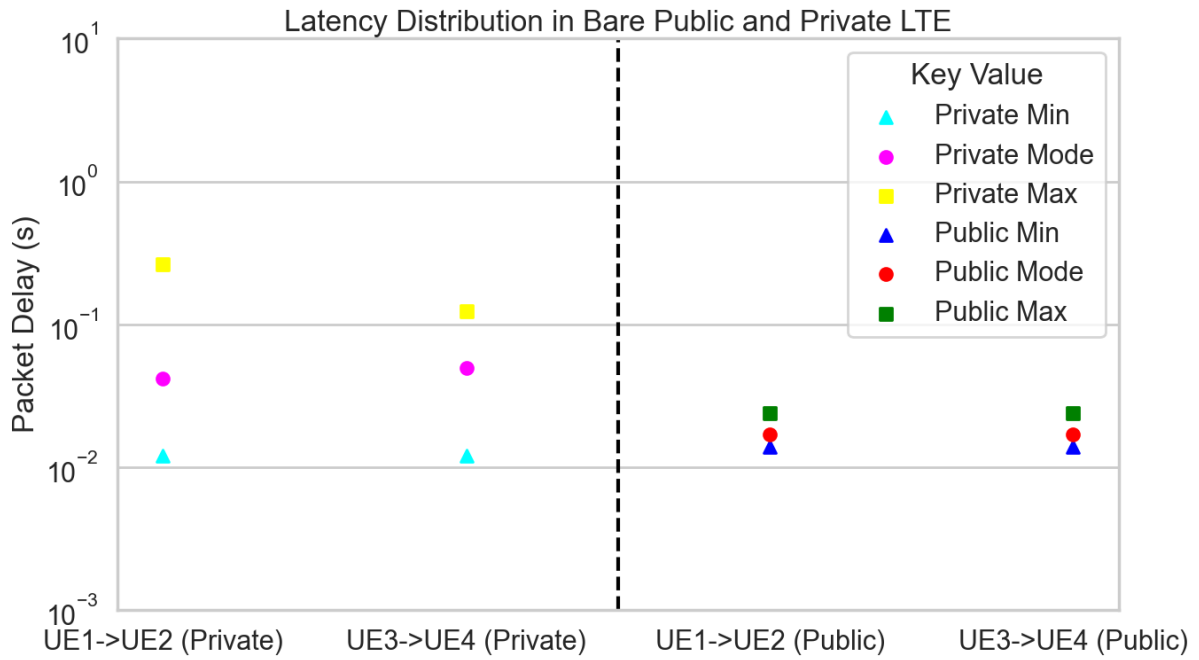


Figure 4.6: Latency Distribution in Bare LTE Networks. The X axis is the flows between UEs, and the Y axis is the delay range with a second of a logarithmic scale. The triangle, circle, and square are the minimum, mode, and maximum delay of each flow, the color distinguishes the type of network, and the dashed line separates each type.

Figures 4-6 show that the public bare LTE network has all its packets with a maximum delay of less than 0.05 seconds. However, the private LTE has more packets spreading up to around 0.28 seconds delay. The mode delay markers indicate that most packets arrive in a public bare LTE network around 0.025 seconds. Similarly, most packets in private bare LTE have a delay of around 0.05 seconds. The private bare LTE network also has a larger maximum delay than the public one.

However, this result is reasonable considering the throughput in both bare LTE networks and our assumption of the capacity of the core network. (Table 4-3 Throughput in Bare LTE Networks) Our work discovered that private LTE has a smaller capacity than public LTE in

simulation in racing scenarios. In this case, when the TCP application adds traffic to the network, the smaller capacity in private LTE requires more time to proceed with these data leading to a large delay range from the packets. In comparison, the larger capacity in public LTE consumes less time to handle them [44] resulting in a smaller day range.

4.6 Summary

This chapter first discusses the flow monitor module in NS-3, whose mechanism is to tag the packet to capture the status of the data flows [39]. The flow monitor calculates lost packets if they are inactive longer than 10 seconds. That might not be the most accurate if the flows counteract more significant latency in an exceptional case. However, it is more efficient to monitor various attributes from each flow using the flow monitor compared with Pcap, ASCII, and traceback functions in NS-3.

The selected metrics for evaluation are throughput, goodput, and latency from flow monitor module [39] in NS-3. They are not direct values from the ".xml" files generated by the flow monitor, but a simple calculation can obtain these metrics based on the equations. (4.1 and 4.2) The latency distribution utilizes the delayhistogram from the flow monitor and the plots use the logarithmic method to display the same scale for all the LTE networks flow, the distribution is based on counts (number of packets) from the delayhistogram in the flow monitor module. Selecting these metrics because they can reflect the performance by throughput and quality by goodput and latency for both traffic patterns in all LTE networks, and these metrics are measurable through flow monitor in NS-3 [40].

Each LTE network has two traffic patterns to distinguish in the throughput plots. Public LTE networks show that the throughput among the flows RT-1, RT-2, and RT-3 is around 260 kB/s as the median. However, the flow BA streaming has around 50 kB/s only in a full traffic pattern. However, the public LTE has a higher throughput close to 300 kB/s from the flows RT-1, RT-2, and RT-3 in light traffic patterns. The flow CB footage improves slightly compared to the full traffic pattern, but the flow BA streaming remains the lowest throughput and longest error bar. The public LTE's goodput has the same trend as their throughput plots in both traffic patterns.

Private LTE has a different trend in throughput from public LTE. It has a lower throughput across all the flows. The flow of CB footage and BA streaming connecting to the public base station has a higher throughput than the flow of RT-1, RT-2, and RT-3, which are attached to a private base station in a full traffic pattern. However, the difference in throughput between flows in the two stations is smaller in light traffic patterns. But the highest throughput is

about 180 kB/s in private LTE networks less than public ones. Interestingly, the flow of BA streaming also has an extremely low throughput in full traffic patterns only. The goodput of the private LTE also remains the same as that of the throughput plots.

As for the latency distribution for both LTE networks. The public LTE has most of its packets filled between 0 second and 0.1 second in all flows regardless of the traffic pattern. The private LTE has a different distribution in that packets are filling in a larger delay range between 0 second to 6 seconds. However, most of the packet in private LTE has a delay between 0 second and 1 second. The public LTE network distributes most of its packets with a smaller delay than the private case.

To discover the root cause of the difference in two LTE networks related to throughput and latency distribution. We conduct the additional simulation called "bare LTE networks" to determine if the measurements of public and private LTE can depend on the capacity of the core network. The results show that the private bare LTE with two base stations connecting to one core network has less throughput and more packets spreading over a larger delay range than the public bare LTE. Which can indicate that a particular capacity of the private LTE networks in whichever simulations we test is smaller than the public one, leading to the queueing of the packets with lower throughput and larger delays [44].

5 Discussion

5.1 Discovery

Multi-Token. We do have some critical observations during the construction of the simulations, including the multiple keys to retrieve the data record from "sample" to "sample-data" table in RACECAR dataset [12]. Moreover, the additional factors affecting results from NS-3 simulation in both LTE models [8].

The conventional case for the RACECAR dataset is that the token of the scene can link to the corresponding logfile and sample record [4]. Which is a general method to identify relevant data, and each token is distinguishable and unique as the primary key. However, this method is unsuitable for "sample" and "sample-data" tables. The way to retrieve all sample-data from a sample token is to use a sample-token and the sensor acting as tokens simultaneously. Because this is the only way to obtain all the sample data based on a specific sample, otherwise the scripts end up with a "key error" message [4]. We appoint out this situation other than investigating the root cause of it.

However, applying two types of tokens is still insufficient. Some of sample-data are using "FRONT" sensor, and others can use "LEFT" or "RIGHT" only. It is still impossible to obtain all the sample-data by just setting one as the 'sensor token'. Which is saying that the extraction scripts have to use a mechanism to detect if one of the sensor values can match the one in sample-data, if not, try with another one. With this technique plus the sample-token only, all the sample-data of a specific sample is fully accessible [4].

Inference of Location in NS-3. Numerous factors can lead to different results in the simulation of NS-3. However, one thing can make a particular link having only 3 to 4 packets to transfer when other links have around 6000 packets [4]. That is because the base stations in the private LTE network are located at the same spots, which should not happen occasionally in real life. Because one base station (radio tower or access point) should serve a group of UEs within specific areas. Nevertheless, this setting does not have a good result in NS-3, it would not benefit some links if multiple base stations stay in the same spots.

Private Spectrum Cellular Network. Our topic uses a private spectrum cellular network to accommodate a racing scenario in the title. But the reason we changed this term to

private LTE in the formal contents is that a private-spectrum cellular network is the specific method of it. Which is also a special case of Private-Public LTE with shared RAN deployment as Chapter 2 introduces. However, when we discuss the methodology and performance in Chapters 3 and 4, we use the term private LTE to indicate that the results are based on this network environment, the private-spectrum cellular network is the method to construct it. In fact, changing any spectrum in NS-3 does not affect any of our measurements unless we change the network architecture. Therefore, we insist on using private LTE to ensure the simulation output is based on this conditions that can affect the results. Moreover, we keep public LTE along to have a comparable baseline to evaluate the performance of private LTE although it does not appear in the title of our work.

5.2 Limitation

Unsupported Z Axis in NS-3 Mobility. The RACECAR dataset stores the translation data with X, Y, and Z coordinates. Unfortunately, the Z-axis does not receive any support from the NS-3.41 version [8]. The Z axis can still affect the simulation measurements because it is one of the coordinates in the mobility module from NS-3. However, our simulations configure the mobility of the nodes with X and Y coordinates. The RACECAR dataset does not specify the laps, the average, or the instant speed that they have for each data sample. We do not take these aspects into our methodology and evaluation either.

Physical Layers. The simulation measurement is not involved with the physical layer, NS-3 does have relevant functions in the physical layer for LTE models [8]. However, we specify some physical parameters including the bandwidth in uplink and downlink, antenna model, and path loss model as other parameters in Chapter 3 [8]. However, the results from our measurements cannot indicate the performance related to any hardware devices. For instance, the measurement varies depending on the locations of base stations, but it is impossible to justify what is the approximate coverage for public LTE or private LTE networks. Or the transmission power, signal-to-noise ratio, etc. Our evaluation focuses on throughput, goodput, and latency where it is above layer 4 because flow monitor [39] can only support the metrics above this layer.

The Earfcn code is an identifier for a range of frequency bands in uplink and downlink [22]. NS-3 can only support the first 30 bands in the LTE network, otherwise, the simulation cannot read Earfcn code and return a zero value on carrier frequency. More importantly, changing the Earfcn code does not affect the measurements by any means. Therefore, this factor does not appear in the table of other parameters in Chapter 3. However, we still use

different frequency bands in two base stations to match the topic.

Limitation of Flow Monitor. The flow monitor module provides a list of metrics to calculate throughput, goodput, and delayhistorgam to analyze the latency distribution. Two features in flow monitor can affect the accuracy of the output, the first one is that the principle to count lost packets is if the packets are not active longer than 10 seconds. It is a reasonable timeout [8], but the timeout of TCP flows mostly varies depending on actual round trip time (RTT) [45]. Which explains that the evaluation does not rely on the lost packet to evaluate the network performance in private and public LTE networks. Moreover, the retransmission packets get counted in sending packets [8], which has a chance to lead to disorder packets to bring negative experience, for instance, lag footage for the audience to watch racing live on their Apple Vision Pro. However, we do not consider the order of the packets in the evaluation. Finally, the flow monitor module does not support any metrics in physical or MAC layer from the NS-3.41 version.

Security and Authentication. Chapter 2 emphasizes the advantage of security in the private LTE network compared to public LTE. However, our work does not cover the security aspects. For instance, when the racing car (UEs) are communicating with pit crews (UEs) through base stations (BSs), it does not get involved with HSS (Home Subscriber Server) and MME (Mobility Management Entry) [6] to authenticate the UEs before data transmission. We concentrate on the fundamental performance assuming each UE node can communicate bidirectionally in every link. However, realistic communication is more complicated than our simulations with additional layers of encryption and authentication.

Core Network in NS-3.41 LTE Model. We found two main limitations of NS-3.41 based on different simulations. The first one is that the capacity core network in the LTE network model might be a constant, because when a private LTE with two base stations is implemented. It possibly split the core network's capacity, leading to lower throughput and a larger delay range for the transmission of the packets. However, the adjustment in datarate of the core network does not change the fact that an LTE network with two base stations is not performing better than just a single base station setup. Therefore, it is possible to speculate that this condition appears due to the rooting setting in NS-3.41 having a fixed capacity for the core network in the LTE model. However, we do not intend to optimize this condition in NS-3 since it is out of our topic scope.

Setting a second core network to provide more capacity is still impossible. However, NS-3.41 does not support more than one core network in a single LTE network simulation. Therefore,

there is no way to assign two network operators in the NS-3.41 version to optimize the performance of the private LTE network with a racing scenario. This question was posted to the NS-3 user group to seek a possible answer but did not have any replies. Our work expects that the NS-3 community would make multiple core network settings possible in future versions.

Communication Protocol in F1 Games. We list a research question about the current communication protocol for motorsport and its challenges in Chapter 1. We attempt to find the relevant academic documents or blogs based on F1 racing. The available information to the public is that F1 does have a dedicated facility [46]. It is responsible for all the internal communication for each team, the technical information of the racing including the statistics of laps and time for each car. It is also the hub for distributing the footage to licensed broadcasters. However, the communication protocol and operation procedure are confidential, and even the layout of this facility is unavailable to the public. Therefore, we claim the first research question is unsolved due to confidentiality. However, F1 is only a subset of the whole motorsport industry. We use F1 as the reference point of our simulation because it represents an advanced level for the capable network technologies considering the vehicle’s speed and the complexity of managing the task of broadcasting.

6 Conclusion

6.1 Conclusion

Based on a racing scenario, we implement simulations for public LTE and private LTE networks. The measurement shows that the private LTE has lower throughput across all the flows than the public LTE network. In addition, the private LTE distributes more packets in the larger delay range up to 6 seconds. However, public LTE has all its packets with delay below 0.1 second. The same trend appears in both full and light traffic patterns. The public LTE always has higher throughput inflows of all racing teams (RT-1, RT-2, RT-3), but it goes the opposite in private LTE where CB footage and BA streaming flows have slightly higher throughput than others.

We appointed five research questions, where the protocol for current racing events to communicate between pit crew and racing driver is unsolved due to its public confidentiality. The method to measure private and public LTE networks is to use NS-3 to run the same iterations of the simulation by each network. The expected results from two LTE networks are that private LTE should perform better, although the actual output does not. The metrics selected to measure the performance of LTE networks are throughput, goodput, and latency distribution (delayhistogram) since the flow monitor module in NS-3 can provide the basic attributes to calculate them. As a result, The non-private spectrum cellular network as a public LTE in the simulation performs with higher throughput from 3 of 5 flows, and packets have lower latency distribution with private-spectrum as a private LTE. However, the public LTE has extremely low throughput in one particular flow. The potential reason is that the only one base station lead to bottleneck of network traffic and the flow BA streaming has longer distance between two nodes.

The scripts for each simulation use the Python bindings API in NS-3, contributing to the community because it only has examples for LTE network models with C++. Moreover, we provide a novel use case for the RACECAR dataset. However, one of the possible reasons for private LTE that cannot have higher throughput and more packets in a lower delay range is the capacity of the core network in NS-3. Therefore, the next step should be making private LTE outperform the public one with the same configuration as our simulations.

6.2 Future Work

NS-3 mmWave Module. Our simulations use NS-3.41 version [25], the newest version until April 2024. However, it can only support 4G (LTE) infrastructures, and an NS-3 millimeter wave module is available [9] for mmWave 5G simulation. Which does not use the same communication mechanism as NS-3.41 has, the more important thing is that NS-3 still does not support 5G New Radio Frequency bands despite version [8]. However, it is still worth examining the different performances in a racing scenario from the NS-3 mmWave module. Because it can produce different results with the same configuration we use.

Authentication. One of the future stages for our simulations should be authentication, especially for the communication between the racing cars and pit crews. Because the internal data transmission in the teams should not allow external users including broadcaster and audience to access it. In our simulations, it is possible to directly transmit the data between an audience and pit crew by simply changing the install node from TCP applications. Therefore, authentication should be essential for a more comprehensive LTE network.

Physical Layer Measurement. From the perspective of a network carrier, it is more valuable for them to know the optimal condition for hardware configuration. For example, what is the best coverage for placing base stations in public and private LTE networks, the power consumption during a racing event, and the signal strength in both networks. With these measurements in hardware level [2], they can better understand how to enable an engaging user experience for racing events and fully take advantage of private LTE networks.

Full Application in RACECAR Dataset. This thesis project only applies few attributes from the RACECAR dataset in nuScene platform [28], but there are still fruitful options including sensors and radars of their vehicles. That can bring the LTE simulation to another level if the simulation can run with more sophisticated parameters and mobility conditions. That can become a part of vehicle-to-vehicle communication in 5G NR applications [47].

Stand-Alone Private LTE Network and Network Slicing. This thesis project only tends to isolate a private network by adding a dedicated base station. However, another way to isolate UEs without adding a new base station is the network slicing [48] by classifying various network resources in the same base station by virtualization. The current NS-3 does not support network slicing, if the future version can utilize this mechanism in our simulations, it will provide additional methods to construct a private LTE network.

Another potential solution to isolate UEs is to deploy a fully stand-alone network for racing drivers and pit crews [20]. The difference between a stand-alone and shared RAN private LTE is that a stand-alone model has a dedicated core network connecting to a private base station for pit crew and racing drivers. The shared RAN still relies on one mutual core network for both base stations. However, NS-3.41 does not support this configuration.

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Between May and June 2024, I used ChatGPT 3.5 to enhance some of my texts when I had trouble expressing my opinions in documentation. This means that my thesis work has ChatGPT involved for writing purposes.

6.4 Availability

All the drafts of the developing files for this thesis and the measurement scripts are here: <https://github.com/liyang19960201/thesis-project-ns3.41.git>.

All the datasets generated are available here: <https://github.com/liyang19960201/Racecar-Dataset-Eval>

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